Control Systems

Previous Year Questions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Control Systems (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
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- 5+ years of teaching experience

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- AIR 19 — GATE (IN)
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- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy

Medium

Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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Contents

Sl.	Topic	Pages
1	Basics of Control Systems	1
2	Block Diagram and Signal Flow Graph	3
3	Time Response Analysis	11
4	Routh's Stability Criterion	21
5	Root Locus	27
6	Polar Plot & Frequency Response of 2nd Order Systems	34
7	Nyquist Plot	38
8	Bode Plot	45
9	State Space Analysis	51
10	Controllers and Compensators	61

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UNIT

I

Basics of Control Systems

Topics

1. Introduction to Control Systems
2. Laplace Transform and Concept of a System
3. Transfer Function
4. Concept of Feedback, OLTF and Sensitivity
5. Poles, Zeros and Final Value Theorem
6. Dominant Pole, Impulse and Step Response

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Unit Snapshot*

Stream: EC	**Questions:** 7	**MCQ:** 6	**MSQ:** 0	**NAT:** 0
1M: 5	**2M:** 2	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 0M(2015)–0M()

**See preface for how Avg Weightage and Range are calculated.*

Q1.15.1 MCQ 2015 1M

Negative feedback in a closed loop control system DOES NOT

A. reduce the overall gain

B. reduce bandwidth

C. improve disturbance rejection

D. reduce sensitivity to parameter variation

Q1.05.1 MCQ 2005 1M

Despite the presence of negative feedback, control systems still have problems of instability because the

A. Components used have nonlinearities

B. Dynamic equations of the subsystems are not known exactly

C. Mathematical analysis involves approximations

D. System has large negative phase angle at high frequencies

Q1.03.1 MCQ 2003 2M

The driving point impedance $Z(s)$ of a network has the pole-zero locations as shown in figure. If $Z(0) = 3$, then $Z(s)$ is

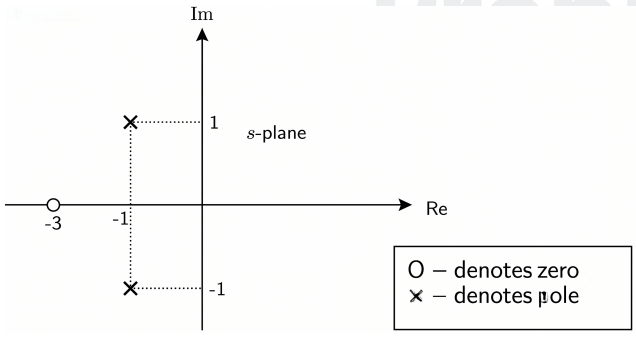

Fig. Q1.03.1

A. $\frac{3(s+3)}{s^2+2s+3}$ **B.** $\frac{2(s+3)}{s^2+2s+2}$

C. $\frac{3(s-3)}{s^2-2s-2}$ **D.** $\frac{2(s-3)}{s^2-2s-3}$

Q1.01.1 MCQ 2001 2M

The open-loop dc gain of a unity negative feedback system with closed-loop transfer function $\frac{s+4}{s^2+7s+13}$ is

A. $\frac{4}{13}$ **B.** $\frac{4}{9}$ **C.** 4 **D.** 13

Q1.98.1 MCQ 1998 1M

The transfer function of a tachometer is of the form

A. Ks **B.** $\frac{K}{s}$ **C.** $\frac{K}{(s+1)}$ **D.** $\frac{K}{s(s+1)}$

Q1.95.1 MCQ 1995 1M

The transfer function of a linear system is the

A. Ratio of the output, $V_0(t)$ and input $V_i(t)$.

B. Ratio of the derivatives of the output and the input.

C. Ratio of the Laplace transform of the output and that of the input with all initial conditions zeroes.

D. None of these.

Q1.94.1 Descriptive 1994 1M

Tachometer feedback in a D.C. position control system enhances stability (T/F) _____

Answer Key

Q1.15.1 **B**	Q1.05.1 **A**	Q1.03.1 **B**
Q1.01.1 **B**	Q1.98.1 **A**	Q1.95.1 **C**
Q1.94.1 **True**		

UNIT

II

Block Diagram and Signal Flow Graph

Topics

1. Block Diagram Reduction Techniques
2. Signal Flow Graphs and Mason's Gain Formula
3. Limitations of Mason's Gain Formula

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Unit Snapshot*

Stream: EC	**Questions:** 27	**MCQ:** 21	**MSQ:** 1	**NAT:** 5
1M: 10	**2M:** 14	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 0M(2016)–2M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

A control system is shown in the Figure. Which option represents the correct transfer function of the system?

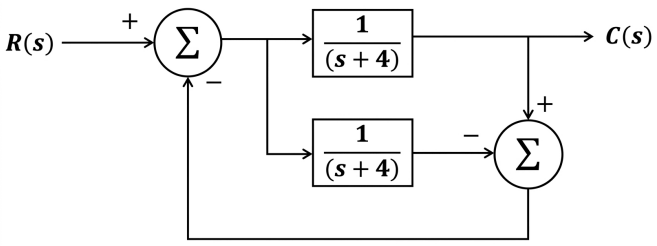


Fig. Q2.26.1

A. $\frac{1}{(s+4)^2}$

B. $\frac{1}{(s+4)}$

C. $\frac{2}{(s+4)}$

D. $\frac{1}{(s^2+8s+17)}$

Q2.25.1 MSQ 2025 2M

Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) this system? Choose the correct option(s).

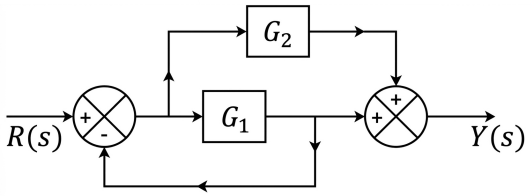
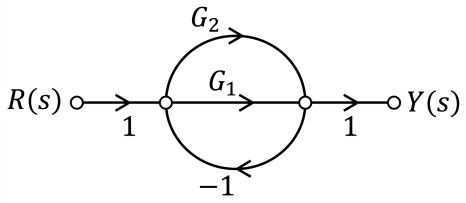
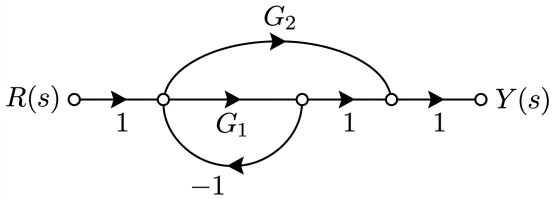


Fig. Q2.25.1

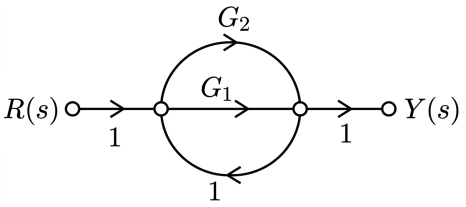
☐ **A.**



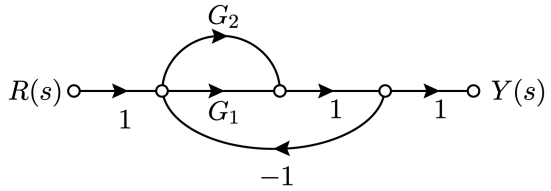
☐ **B.**



☐ **C.**



☐ **D.**



Q2.23.1 MCQ 2023 2M

In the following block diagram, $R(s)$ and $D(s)$ are two inputs. The output $Y(s)$ is expressed as

$$Y(s) = G_1(s)R(s) + G_2(s)D(s).$$

$G_1(s)$ and $G_2(s)$ are given by

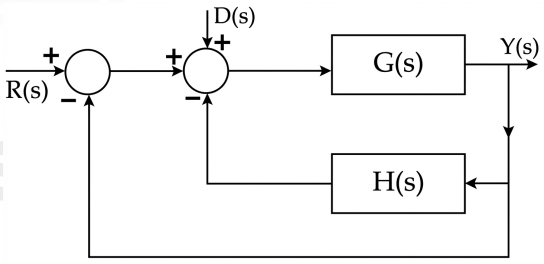


Fig. Q2.23.1

A. $G_1(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$, $G_2(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$

B. $G_1(s) = \frac{G(s)}{1+G(s)+H(s)}$, $G_2(s) = \frac{G(s)}{1+G(s)+H(s)}$

C. $G_1(s) = \frac{G(s)}{1+G(s)+H(s)}$, $G_2(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$

D. $G_1(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$, $G_2(s) = \frac{G(s)}{1+G(s)+H(s)}$

Q2.21.1 MCQ 2021 1M

The block diagram of a feedback control system is shown in the figure.

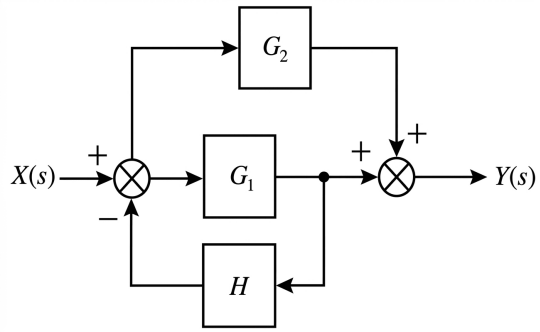


Fig. Q2.21.1

The transfer function $\frac{Y(s)}{X(s)}$ of the system is

A. $\frac{G_1 + G_2 + G_1G_2H}{1 + G_1H}$ **B.** $\frac{G_1 + G_2}{1 + G_1H + G_2H}$

C. $\frac{G_1 + G_2}{1 + G_1H}$ **D.** $\frac{G_1 + G_2 + G_1G_2H}{1 + G_1H + G_2H}$

Q2.19.1 MCQ 2019 2M

The block diagram of a system is illustrated in the figure shown, where $X(s)$ is the input and $Y(s)$ is the output. The transfer function $H(s) = \frac{Y(s)}{X(s)}$ is

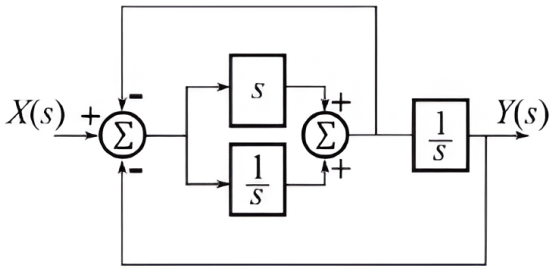


Fig. Q2.19.1

A. $H(s) = \frac{s^2 + 1}{s^3 + s^2 + s + 1}$ **B.** $H(s) = \frac{s^2 + 1}{s^3 + 2s^2 + s + 1}$

C. $H(s) = \frac{s + 1}{s^2 + s + 1}$ **D.** $H(s) = \frac{s^2 + 1}{2s^2 + 1}$

Q2.17.1 NAT 2017 1M

For the system shown in the figure, $Y(s)/X(s) =$ _____

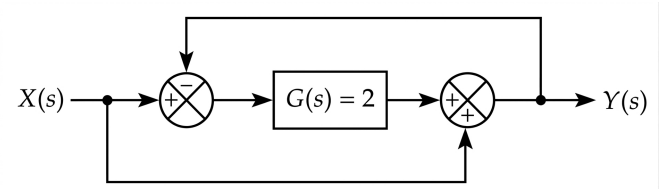


Fig. Q2.17.1

Q2.16.1 MCQ 2016 1M

The block diagram of a feedback control system is shown in the figure. The overall closed-loop gain G of the system is

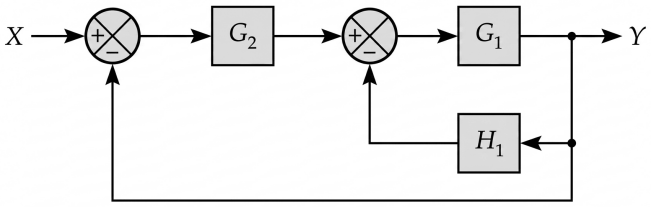


Fig. Q2.16.1

A. $G = \frac{G_1G_2}{1 + G_1H_1}$ **B.** $G = \frac{G_1G_2}{1 + G_1G_2 + G_1H_1}$

C. $G = \frac{G_1G_2}{1 + G_1G_2H_1}$ **D.** $G = \frac{G_1G_2}{1 + G_1G_2 + G_1G_2H_1}$

Q2.15.1 MCQ 2015 1M

For the signal flow graph shown in the figure, the value of $C(s)/R(s)$ is

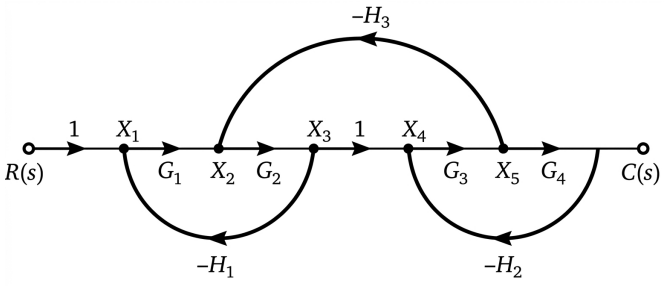


Fig. Q2.15.1

A. $\frac{G_1G_2G_3G_4}{1 - G_1G_2H_1 - G_3G_4H_2 - G_2G_3H_3 + G_1G_2G_3G_4H_1H_2}$

B. $\dfrac{G_1G_2G_3G_4}{1+G_1G_2H_1+G_3G_4H_2+G_2G_3H_3+G_1G_2G_3G_4H_1H_2}$

C. $\dfrac{1}{1+G_1G_2H_1+G_3G_4H_2+G_2G_3H_3+G_1G_2G_3G_4H_1H_2}$

D. $\dfrac{1}{1-G_1G_2H_1-G_3G_4H_2-G_2G_3H_3+G_1G_2G_3G_4H_1H_2}$

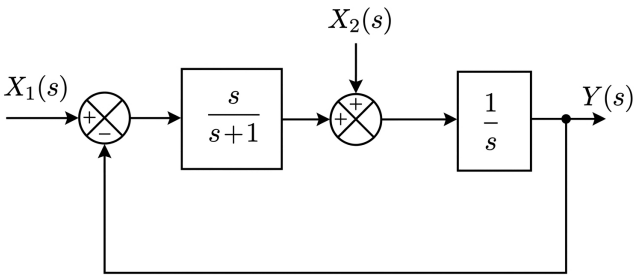


Fig. Q2.14.1

When $X_1(s) = 0$, the transfer function $\dfrac{Y(s)}{X_2(s)}$ is

A. $\dfrac{s+1}{s^2}$ **B.** $\dfrac{1}{s+1}$ **C.** $\dfrac{s+2}{s(s+1)}$ **D.** $\dfrac{s+1}{s(s+2)}$

Q2.15.2 MCQ 2015 1M

By performing cascading and/or summing/differencing operations using transfer function blocks $G_1(s)$ and $G_2(s)$, one CANNOT realize a transfer function of the form

A. $G_1(s)G_2(s)$ **B.** $\dfrac{G_1(s)}{G_2(s)}$

C. $G_1(s)\left(\dfrac{1}{G_1(s)}+G_2(s)\right)$ **D.** $G_1(s)\left(\dfrac{1}{G_1(s)}-G_2(s)\right)$

Q2.15.3 NAT 2015 2M

The position control of a DC servo-motor is given in the figure. The values of the parameters are $K_T = 1$ N-m/A, $R_a = 1\Omega$, $L_a = 0.1$H, $J = 5$kg-m^2, $B = 1$ N-m/(rad/sec) and $K_b = 1$V/(rad/sec).

The steady-state position response (in radians) due to unit impulse disturbance torque T_d is

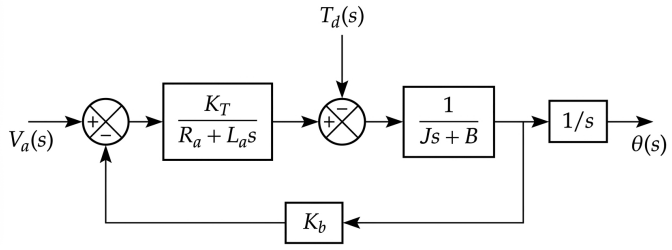


Fig. Q2.15.3

Q2.14.1 MCQ 2014 1M

For the following system

Q2.14.2 MCQ 2014 1M

Consider the following block diagram in the figure. The transfer function $C(s)/R(s)$ is

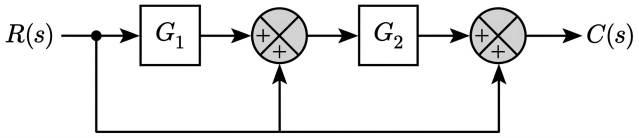


Fig. Q2.14.2

A. $\dfrac{G_1G_2}{1+G_1G_2}$ **B.** $G_1G_2+G_1+1$

C. $G_1G_2+G_2+1$ **D.** $\dfrac{G_1}{1+G_1G_2}$

Q2.13.1 MCQ 2013 2M

The signal flow graph for a system is given below. The transfer function $Y(s)/U(s)$ for this system is

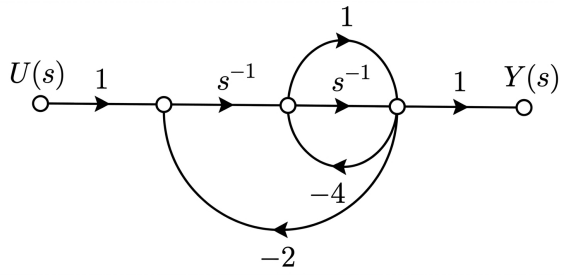


Fig. Q2.13.1

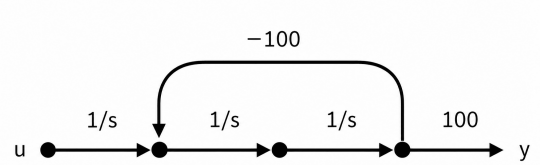
A. $\dfrac{s+1}{5s^2+6s+2}$

B. $\dfrac{s+1}{s^2+6s+2}$

C. $\dfrac{s+1}{s^2+4s+2}$

D. $\dfrac{1}{5s^2+6s+2}$

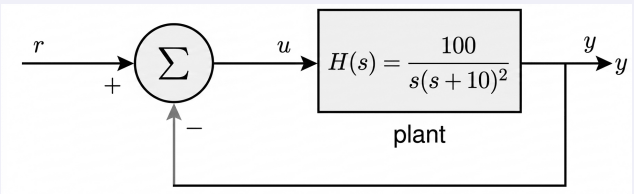
D.



Common Data Questions

Common Data for Questions Q2.11.1 and Q6.11.1:

The input-output transfer function of a plant $H(s) = \dfrac{100}{s(s+10)^2}$. The plant is placed in a unity negative feedback configuration as shown in the figure below.

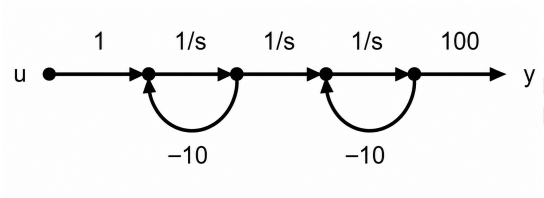


The 2nd Question is given in Unit VI (Q6.11.1)

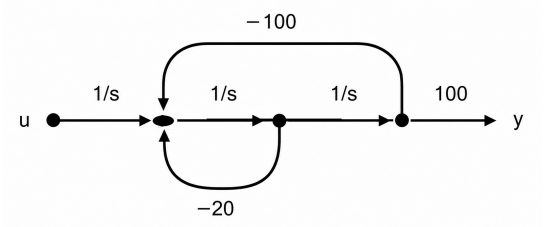
Q2.11.1 MCQ 2011 2M

The signal flow graph that DOES NOT model the plant transfer function $H(s)$ is

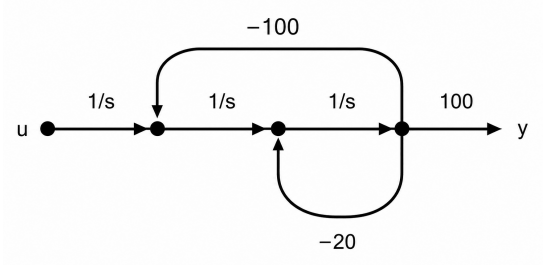
A.



B.



C.



Q2.04.1 MCQ 2004 2M

Consider the signal flow graph shown in figure. The gain $\dfrac{x_5}{x_1}$ is

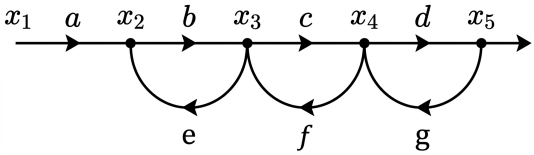


Fig. Q2.04.1

A. $\dfrac{1-(be+cf+dg)}{abcd}$

B. $\dfrac{bedg}{1-(be+cf+dg)}$

C. $\dfrac{abcd}{1-(be+cf+dg)+bedg}$

D. $\dfrac{1-(be+cf+dg)+bedg}{abcd}$

Q2.03.1 MCQ 2003 2M

The signal flow graph of a system is shown in figure. The transfer Function $\dfrac{C(s)}{R(s)}$ of the system is

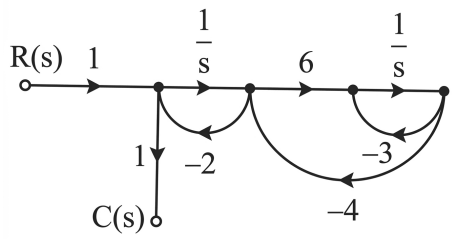


Fig. Q2.03.1

A. $\dfrac{6}{s^2+29s+6}$ **B.** $\dfrac{6s}{s^2+29s+6}$

C. $\dfrac{s(s+2)}{s^2+29s+6}$ **D.** $\dfrac{s(s+27)}{s^2+29s+6}$

Q2.01.1 MCQ 2001 1M

The equivalent of the block diagram in Figure-Q2.01.1 is given as

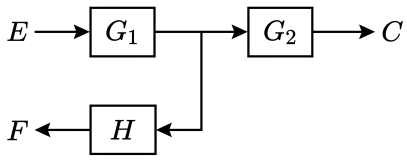
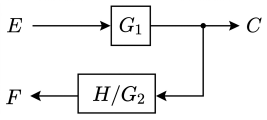
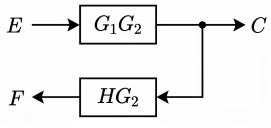
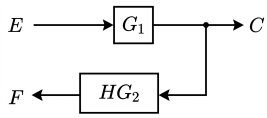
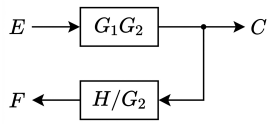


Fig. Q2.01.1

A.


B.


C.


D.


Q2.01.2 MCQ 2001 2M

An electrical system and its signal-flow graph representation are shown in figure (a) and (b) respectively. The values of G_2 and H, respectively, are

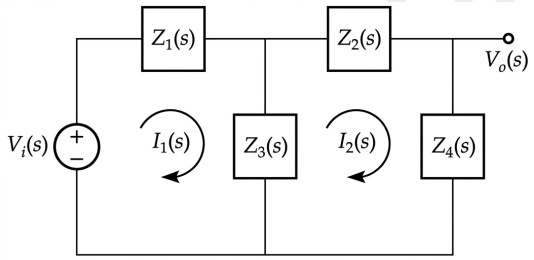


Fig. (a)

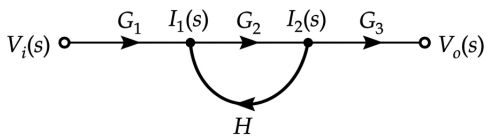


Fig. (b)

Fig. Q2.01.2

A. $\dfrac{Z_3(s)}{Z_2(s)+Z_3(s)+Z_4(s)}, \dfrac{-Z_3(s)}{Z_1(s)+Z_3(s)}$

B. $\dfrac{-Z_3(s)}{Z_2(s)-Z_3(s)+Z_4(s)}, \dfrac{-Z_3(s)}{Z_1(s)+Z_3(s)}$

C. $\dfrac{Z_3(s)}{Z_2(s)+Z_3(s)+Z_4(s)}, \dfrac{Z_3(s)}{Z_1(s)+Z_3(s)}$

D. $\dfrac{-Z_3(s)}{Z_2(s)-Z_3(s)+Z_4(s)}, \dfrac{Z_3(s)}{Z_1(s)+Z_3(s)}$

Q2.01.3 NAT 2001 5M

A feedback control system is shown in figure

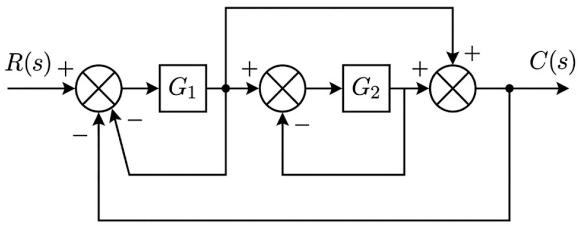


Fig. Q2.01.3

(a) Draw the signal-flow graph that represents the system.

(b) Find the total number of loops in the graph and determine the loop-gains of all the loops.

(c) Find the number of all possible combinations of non-touching loops taken two at a time.

(d) Determine the transfer function of the system using the signal-flow graph.

Q2.98.1 NAT 1998 5M

Draw a signal flow graph for the following set of algebraic equations:

$$y_2 = ay_1 - gy_3$$

$$y_3 = ey_2 + cy_4$$

$$y_4 = by_2 - dy_4$$

Hence, find the gains $\dfrac{y_2}{y_1}$ and $\dfrac{y_3}{y_1}$

Q2.97.1 MCQ 1997 2M

In the signal flow graph of Fig. Y/X equals

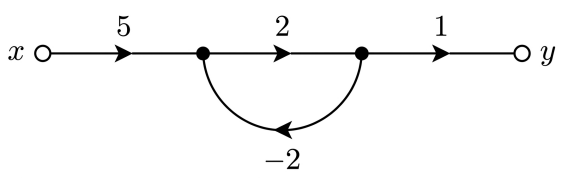


Fig. Q2.97.1

A. 3 **B.** 5/2 **C.** 2 **D.** None of the above

Q2.95.1 MCQ 1995 1M

Signal flow graph is used to find

A. stability of the system

B. controllability of the system

C. transfer function of the system

D. poles of the system

Q2.94.1 NAT 1994 5M

Reduce the signal flow graph shown in fig. below, to obtain another graph which does not contain the node e_5. (Also, remove any self-loop from the resulting graph)

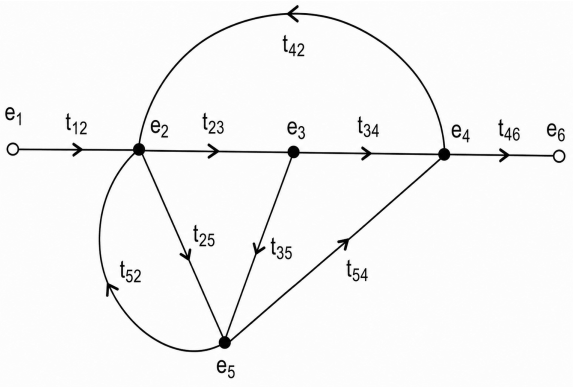


Fig. Q2.94.1

Q2.91.1 MCQ 1991 2M

In the signal flow graph of fig. the gain c/r will be

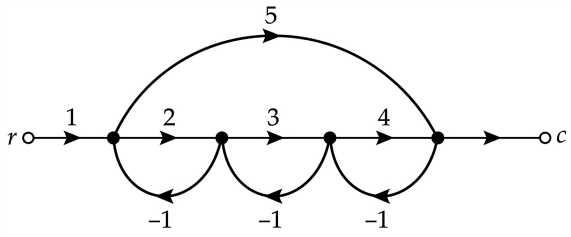


Fig. Q2.91.1

A. 11/9 **B.** 22/15 **C.** 24/23 **D.** 44/23

Q2.89.1 MCQ 1989 2M

The C/R for the signal flow graph in fig. is:

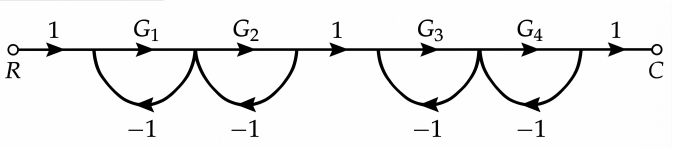


Fig. Q2.89.1

A. $\dfrac{G_1G_2G_3G_4}{(1+G_1G_2)(1+G_3G_4)}$

B. $\dfrac{G_1G_2G_3G_4}{(1+G_1+G_2+G_1G_2)(1+G_3+G_4+G_3G_4)}$

C. $\dfrac{G_1G_2G_3G_4}{(1+G_1+G_2)(1+G_3+G_4)}$

D. $\dfrac{G_1G_2G_3G_4}{(1+G_1+G_2+G_3+G_4)}$

Q2.87.1 MCQ 1987 2M

For the system shown in figure the transfer function $C(s)/R(s)$ is equal to

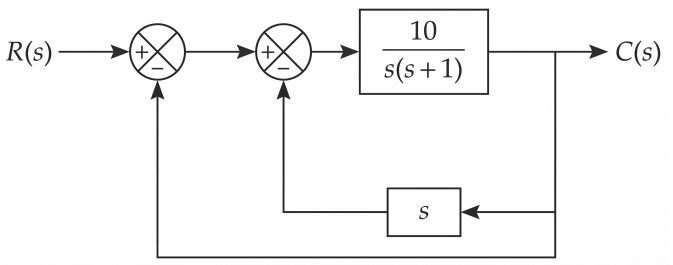


Fig. Q2.87.1

A. $\frac{10}{s^2 + s + 10}$ **B.** $\frac{10}{s^2 + 11s + 10}$

C. $\frac{10}{s^2 + 9s + 10}$ **D.** $\frac{10}{s^2 + 2s + 10}$

Q2.87.2 MCQ 1987 2M

In the signal flow graph shown in fig, $X_2 = TX_1$, where T is equal to

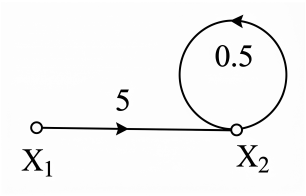


Fig. Q2.87.2

A. 2.5 **B.** 5 **C.** 5.5 **D.** 10

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q2.26.1 B	Q2.25.1 B	Q2.23.1 A
Q2.21.1 C	Q2.19.1 B	Q2.17.1 0.95 to 1.05
Q2.16.1 B	Q2.15.1 B	Q2.15.2 B
Q2.15.3 -0.51 to -0.49	Q2.14.1 D	Q2.14.2 C
Q2.13.1 A	Q2.11.1 D	Q2.04.1 C
Q2.03.1 D	Q2.01.1 D	Q2.01.2 C
Q2.01.3 *	Q2.98.1 *	Q2.97.1 C
Q2.95.1 C	Q2.94.1 *	Q2.91.1 D
Q2.89.1 C	Q2.87.1 B	Q2.87.2 D

UNIT

III

Time Response Analysis

Topics

1. Impulse, Step, Ramp and Parabolic Response
2. Concept of Damping in 2nd Order Systems
3. Complex Frequency and Damping in RLC Circuits
4. Step Response of 2nd Order Systems
5. Time Domain Performance Parameters
6. Steady State Error — Unity and Non-Unity Feedback
7. Error Constants and System Types

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 52	**MCQ:** 28	**MSQ:** 1	**NAT:** 18
1M: 15	**2M:** 25	**Descriptive:** 12	**Avg Weightage** ~2M	**Range:** 0M(2015)–4M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q3.24.1 MCQ 2024 1M

In the feedback control system shown in the figure below $G(s) = \dfrac{6}{s(s+1)(s+2)}$.

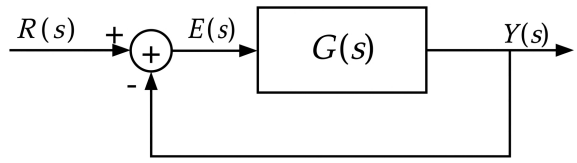


Fig. Q3.24.1

$R(s)$, $Y(s)$, and $E(s)$ are the Laplace transforms of $r(t)$, $y(t)$, and $e(t)$, respectively. If the input $r(t)$ is a unit step function, then _____.

A. $\lim_{t\to\infty} e(t) = 0$

B. $\lim_{t\to\infty} e(t) = \dfrac{1}{3}$

C. $\lim_{t\to\infty} e(t) = \dfrac{1}{4}$

D. $\lim_{t\to\infty} e(t)$ does not exist, $e(t)$ is oscillatory

Q3.23.1 MCQ 2023 2M

A closed loop system is shown in the figure where $k > 0$ and $\alpha > 0$. The steady state error due to a ramp input ($R(s) = \alpha/s^2$) is given by

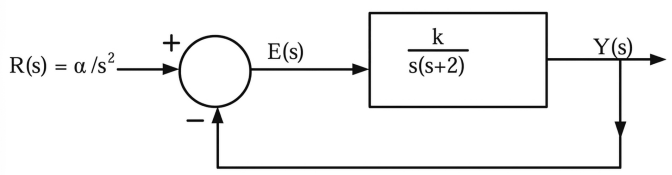


Fig. Q3.23.1

A. $\dfrac{2\alpha}{k}$ **B.** $\dfrac{\alpha}{k}$ **C.** $\dfrac{\alpha}{2k}$ **D.** $\dfrac{\alpha}{4k}$

Q3.22.1 MSQ 2022 2M

Two linear time-invariant systems with transfer functions

$$G_1(s) = \frac{10}{s^2 + s + 1} \text{ and } G_2(s) = \frac{10}{s^2 + s\sqrt{10} + 10}$$

have unit step responses $y_1(t)$ and $y_2(t)$, respectively. Which of the following statements is/are true?

☐ **A.** $y_1(t)$ and $y_2(t)$ have the same percentage peak overshoot.

☐ **B.** $y_1(t)$ and $y_2(t)$ have the same steady-state value.

☐ **C.** $y_1(t)$ and $y_2(t)$ have the same damped frequency of oscillation.

☐ **D.** $y_1(t)$ and $y_2(t)$ have the same 2% settling time.

Q3.22.2 NAT 2022 2M

The block diagram of a closed-loop control system is shown in the figure. $R(s), Y(s)$, and $D(s)$ are the Laplace transforms of the time-domain signals $r(t)$, $y(t)$, and $d(t)$, respectively. Let the error signal be defined as $e(t) = r(t) - y(t)$. Assuming the reference input $r(t) = 0$ for all t, the steady-state error $e(\infty)$, due to a unit step disturbance $d(t)$, is

_____

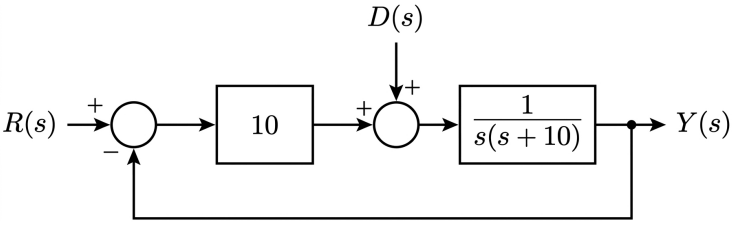


Fig. Q3.22.2

Q3.20.1 NAT 2020 2M

Consider the following closed loop control system

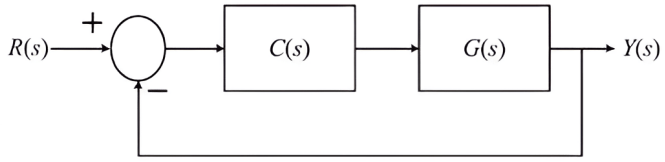


Fig. Q3.20.1

where $G(s) = \dfrac{1}{s(s+1)}$ and $C(s) = K\dfrac{s+1}{s+3}$. If the steady state error for a unit ramp input is 0.1, then the value of K is

_____

Q3.19.1 MCQ 2019 2M

Consider a causal second-order system with the transfer function

$$G(s) = \frac{1}{1 + 2s + s^2}$$

with a unit-step $R(s) = \frac{1}{s}$ as an input. Let $C(s)$ be the corresponding output. The time taken by the system output $c(t)$ to reach 94% of its steady-state value $\lim_{t\to\infty} c(t)$, rounded off to two decimal places, is

A. 5.25 **B.** 4.50 **C.** 3.89 **D.** 2.81

Q3.17.1 NAT 2017 1M

The open-loop transfer function,

$$G(s) = \frac{(s+1)}{s^p(s+2)(s+3)}$$

where, p is an integer, is connected in unity feedback configuration as shown in the figure.

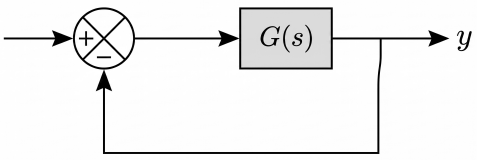


Fig. Q3.17.1

Given that the steady-state error is zero for unit step input and is 6 for unit ramp input, the value of the parameter p is

_____

Q3.16.1 NAT 2016 1M

The response of the system $G(s) = \frac{s-2}{(s+1)(s+3)}$ to the unit step input $u(t)$ is $y(t)$. The value of $\frac{dy}{dt}$ at $t = 0^+$ is

_____

Q3.16.2 NAT 2016 2M

The open-loop transfer function of a unity-feedback control system is given by

$$G(s) = \frac{K}{s(s+2)}$$

For the peak overshoot of the closed-loop system to a unit step input to be 10%, the value of K is

_____

Q3.16.3 NAT 2016 2M

In the feedback system shown below

$$G(S) = \frac{1}{(s^2 + 2s)}$$

The step response of the closed-loop system should have minimum settling time and have no overshoot.

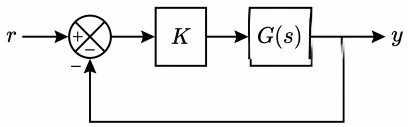


Fig. Q3.16.3

The required value of gain k to achieve this is

_____

Q3.15.1 NAT 2015 1M

A unity negative feedback system has an open-loop transfer function $G(s) = \frac{K}{s(s+10)}$. The gain K for the system to have a damping ratio of 0.25 is

_____

Q3.14.1 NAT 2014 1M

The natural frequency of an undamped second-order system is 40 rad/s. If the system is damped with a damping ratio 0.3, the damped natural frequency in rad/s is

_____

Q3.14.2 MCQ 2014 1M

For the second order closed – loop system shown in the figure, the natural frequency (in rad/s) is

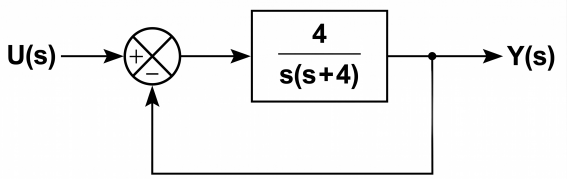


Fig. Q3.14.2

A. 16 **B.** 4 **C.** 2 **D.** 1

Q3.14.3 NAT 2014 2M

The steady state error of the system shown in the figure for a unit step input is

_____

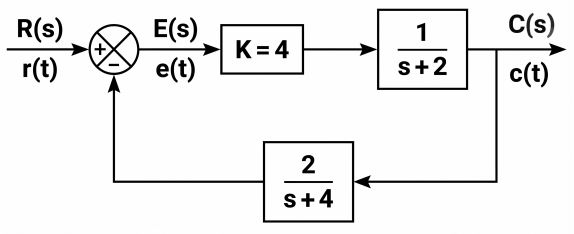


Fig. Q3.14.3

Q3.13.1 MCQ 2013 2M

The open-loop transfer function of a dc motor is given as $\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s}$. When connected in feedback as shown below, the approximate value of K_a that will reduce the time constant of the closed loop system by one hundred times as compared to that of the open-loop system is

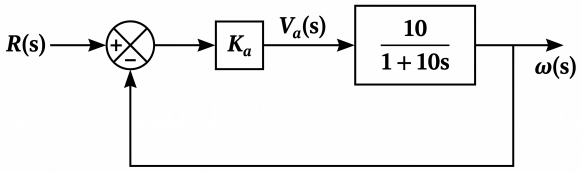


Fig. Q3.13.1

A. 1 **B.** 5 **C.** 10 **D.** 100

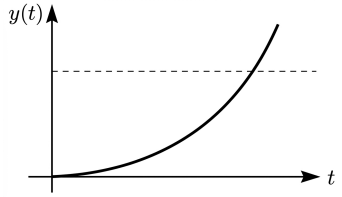
Q3.11.1 MCQ 2011 1M

The differential equation

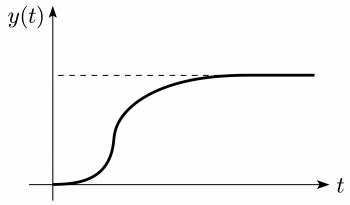
$$100\frac{d^2y}{dt^2} - 20\frac{dy}{dt} + y = x(t)$$

describes a system with an input $x(t)$ and an output $y(t)$. The system, which is initially relaxed, is excited by a unit step input. The output $y(t)$ can be represented by the waveform

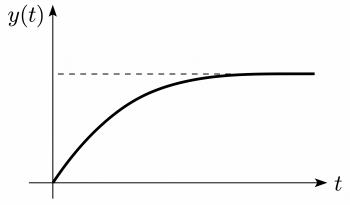
A.



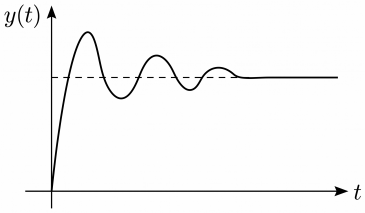
B.



C.



D.



Q3.09.1 MCQ 2009 2M

The unit step response of an underdamped second order system has steady value of −2. Which one of the following transfer functions has these properties?

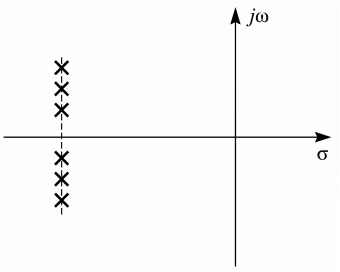
A. $\frac{-2.24}{s^2 + 2.59s + 1.12}$ **B.** $\frac{-3.82}{s^2 + 1.91s + 1.91}$

C. $\frac{-2.24}{s^2 - 2.59s + 1.12}$ **D.** $\frac{-3.82}{s^2 - 1.91s + 1.91}$

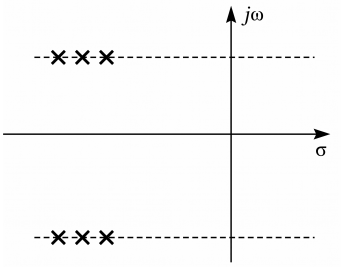
Q3.08.1 MCQ 2008 1M

Step responses of a set of three second-order underdamped systems all have the same percentage overshoot. Which of the following diagrams represents the poles of three systems?

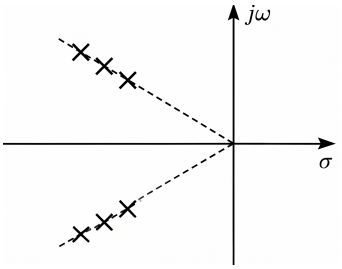
A.



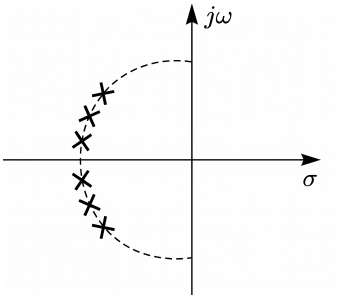
B.



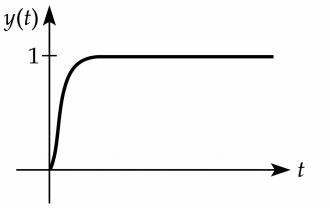
C.



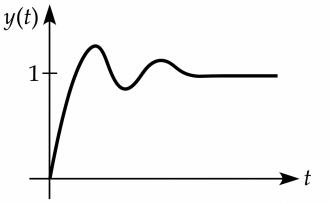
D.



1.



2.



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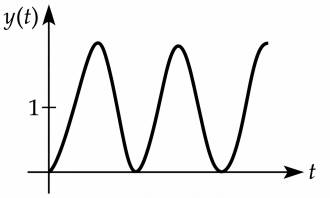

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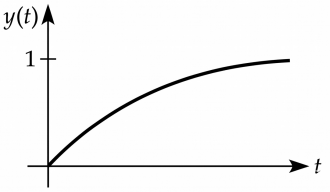


Fig. Q3.08.2

Q3.08.2 MCQ 2008 2M

Group I lists a set of four transfer functions. Group II gives a list of possible step responses $y(t)$. Match the step responses with the corresponding transfer functions.

Group - I

$$P = \frac{25}{s^2 + 25}; \qquad Q = \frac{36}{s^2 + 20s + 36}$$

$$R = \frac{36}{s^2 + 12s + 36}; \qquad S = \frac{49}{s^2 + 7s + 49}$$

Group - II

A.

P	Q	R	S
3	1	4	2

B.

P	Q	R	S
3	2	4	1

C.

P	Q	R	S
2	1	4	3

D.

P	Q	R	S
3	4	1	2

Q3.08.3 MCQ 2008 2M

A linear, time-invariant, causal continuous time system has a rational transfer function with simple poles at $s = -2$ and $s = -4$, and one simple zero at $s = -1$. A unit step $u(t)$ is applied at the input of the system. At steady state, the output has constant value of 1. The impulse response of the system is

A. $[\exp(-2t) + \exp(-4t)]u(t)$

B. $[-4\exp(-2t) + 12\exp(-4t) - \exp(-t)]u(t)$

C. $[-4\exp(-2t) + 12\exp(-4t)]u(t)$

D. $[-0.5\exp(-2t) + 1.5\exp(-4t)]u(t)$

Q3.07.1 MCQ 2007 2M

The transfer function of a plant is

$$T(s) = \frac{5}{(s+5)(s^2+s+1)}$$

The second-order approximation of $T(s)$ using dominant pole concept is

A. $\frac{1}{(s+5)(s+1)}$ **B.** $\frac{5}{(s+5)(s+1)}$

C. $\frac{5}{(s^2+s+1)}$ **D.** $\frac{1}{(s^2+s+1)}$

Q3.05.1 MCQ 2005 2M

A ramp input applied to an unity feedback system results in 5% steady state error. The type number and zero frequency gain of the system are respectively

A. 1 and 20 **B.** 0 and 20

C. 0 and $\frac{1}{20}$ **D.** 1 and $\frac{1}{20}$

Q3.05.2 MCQ 2005 2M

In the derivation of expression for peak percent overshoot

$$M_p = \exp\left(\frac{-\pi\xi}{\sqrt{1-\xi^2}}\right) \times 100\%$$

Which one of the following conditions is NOT required?

A. System is linear and time invariant

B. The system transfer function has a pair of complex conjugate poles and no zeroes

C. There is no transportation delay in the system

D. The system has zero initial conditions

Q3.04.1 MCQ 2004 2M

A causal system having the transfer function $G(s) = \frac{1}{s+2}$ is excited with $10u(t)$. The time at which the output reaches 99% of its steady state value is

A. 2.7 sec **B.** 2.5 sec **C.** 2.3 sec **D.** 2.1 sec

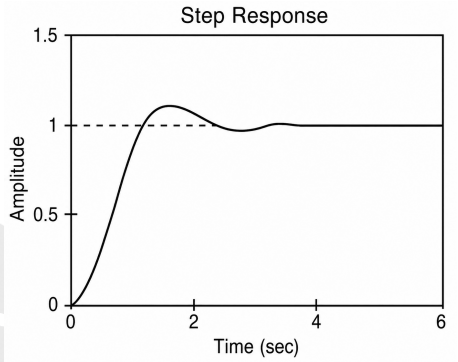
Q3.03.1 MCQ 2003 2M

A second-order system has the transfer function

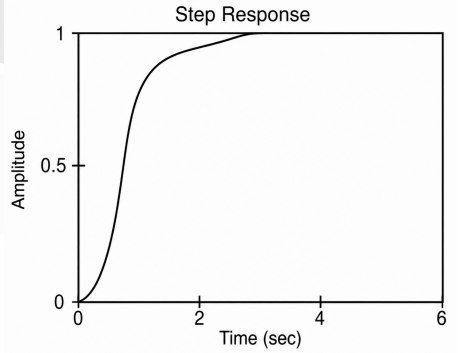
$$\frac{C(s)}{R(s)} = \frac{4}{s^2+4s+4}$$

. With $r(t)$ as the unit-step function, the response $c(t)$ of the system is represented by

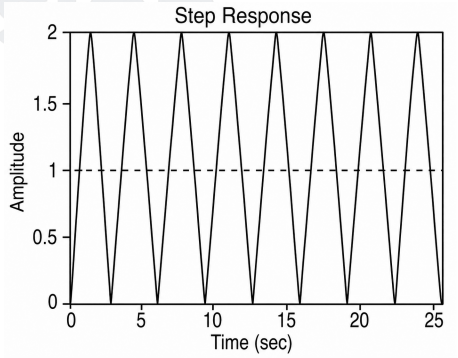
A.



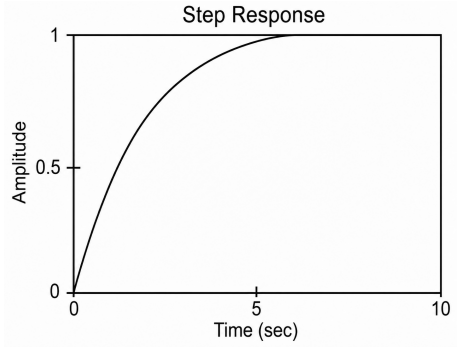
B.



C.



D.



Q3.02.1 MCQ 2002 1M

Consider a system with the transfer function

$$G(s) = \frac{s+6}{Ks^2 + s + 6}$$

Its damping ratio will be 0.5 when the value of K is

A. 2/6 **B.** 3 **C.** 1/6 **D.** 6

Q3.02.2 MCQ 2002 2M

The transfer function of a system is

$$G(s) = \frac{100}{(s+1)(s+100)}$$

For a unit-step input to the system the approximate settling time for 2% criterion is

A. 100 sec **B.** 4 sec **C.** 1 sec **D.** 0.01 sec

Q3.01.1 MCQ 2001 1M

If the characteristic equation of a closed-loop system is $s^2 + 2s + 2 = 0$, then the system is

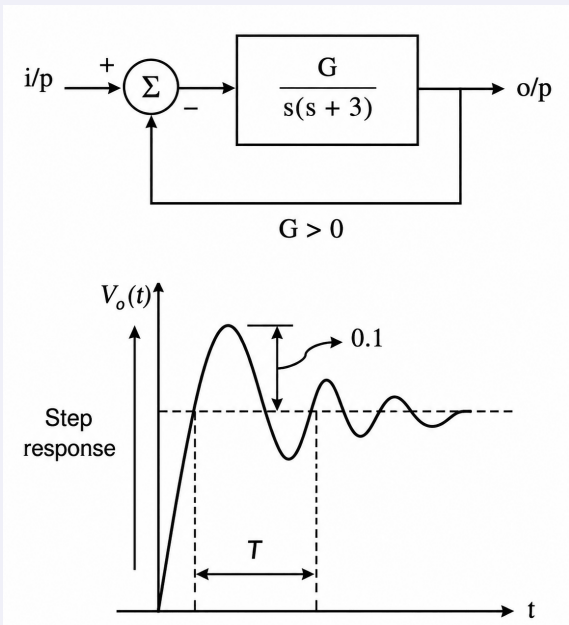
A. Overdamped **B.** Critically damped
C. Underdamped **D.** Undamped

Common Data Questions

The next 3 questions share the data below.

Common data for Q3.00.1, Q3.00.2 and Q3.00.3

The block diagram of a feedback system is shown in the figure.



Q3.00.1 NAT 2000 5M

Find the closed loop transfer function.

Q3.00.2 NAT 2000 5M

Find the minimum value of G for which the step response of the system would exhibit an overshoot, as shown in figure.

Q3.00.3 NAT 2000 5M

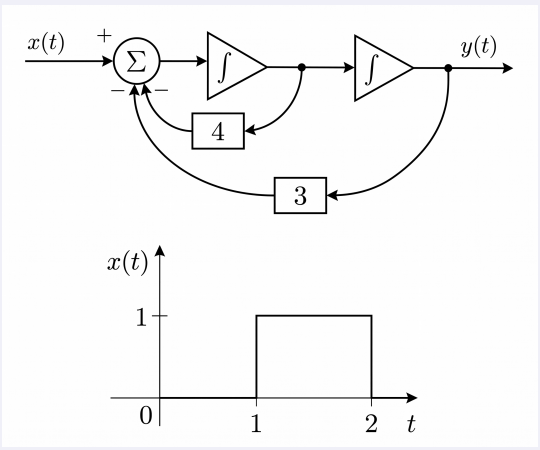
For G equal to twice this minimum value, find the time period T indicated in the figure.

Common Data Questions

The next 3 questions share the data below.

Common data for Q3.00.4, Q3.00.5 and Q3.00.6

For the linear, time-invariant system whose block diagram is shown in figure with input $x(t)$ and output $y(t)$,



Q3.00.4 NAT | 2000 | 8M

Find the transfer function.

Q3.00.5 NAT | 2000 | 8M

Find the step response of the system. [find $y(t)$ when $x(t)$ is a unit step function and the initial conditions are zero]

Q3.00.6 NAT | 2000 | 8M

Find $y(t)$, if $x(t)$ is as shown in figure and the initial conditions are zero.

Q3.99.1 MCQ | 1999 | 1M

For a second order system with the closed-loop transfer function, $T(s) = \dfrac{9}{s^2 + 4s + 9}$ the settling time for 2% band (in seconds) is

A. 1.5 **B.** 2.0 **C.** 3.0 **D.** 4.0

Q3.99.2 MCQ | 1999 | 2M

If the closed-loop transfer function $T(s)$ of a unity negative feedback system is given by

$$T(s) = \frac{a_{n-1}s + a_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n}$$

then the steady-state error for a unit ramp input is

A. $\dfrac{a_n}{a_{n-1}}$ **B.** $\dfrac{a_n}{a_{n-2}}$ **C.** $\dfrac{a_{n-2}}{a_{n-1}}$ **D.** zero

Q3.98.1 MCQ | 1998 | 1M

Consider a feedback control system with loop transfer function

$$G(s)H(s) = \frac{K(1 + 0.5s)}{s(1 + s)(1 + 2s)}$$

The type of the closed loop system is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q3.98.2 MCQ | 1998 | 1M

Consider a unity feedback control system with open-loop transfer function $G(s) = \dfrac{K}{s(s + 1)}$. The steady state error of the system due to a unit step input is

A. zero **B.** K **C.** $1/K$ **D.** infinite

Q3.98.3 Descriptive | 1998 | 10M

Consider the system shown in Fig. Determine the value of 'a' such that the damping ratio is 0.5. Also obtain the value of the rise time 't_r' and maximum overshoot 'M_P' in its step response.

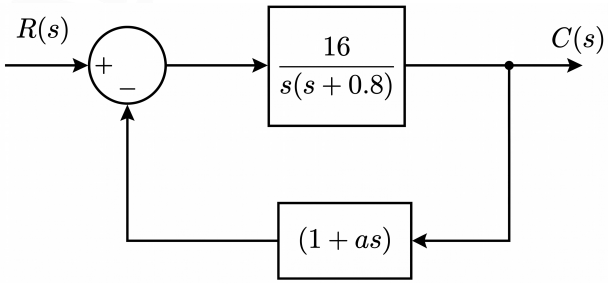


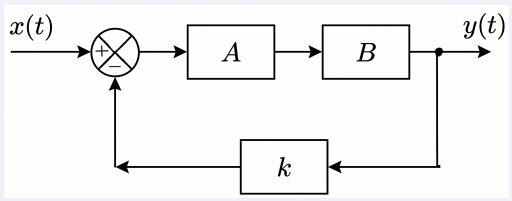
Fig. Q3.98.3

Common Data Questions

The next 3 questions share the data below.

Common data for Q3.97.1, Q3.97.2 and Q3.97.3

Following fig. shows the block diagram representation of a control system. The system in block A has an impulse response $h_A(t) = e^{-t}u(t)$. The system in block B has an impulse response $h_B(t) = e^{-2t}u(t)$. The block 'k' is an amplifier by a factor k. For the over all system the input is $x(t)$ and output $y(t)$.



Q3.97.1 NAT 1997 5M

Find the transfer function $\frac{y(s)}{x(s)}$ when $k = 1$.

Q3.97.2 NAT 1997 5M

Find the impulse response when $k = 0$.

Q3.97.3 NAT 1997 5M

Find the values of k for which the system becomes unstable.

Q3.95.1 MCQ 1995 1M

The step error coefficient of a system $G(s) = \frac{1}{(s+6)(s+1)}$ with unity feedback is

A. $\frac{1}{6}$ **B.** ∞ **C.** 0 **D.** 1

Q3.95.2 MCQ 1995 1M

For a second order system, damping ratio (ξ), is $0 < \xi < 1$, then the roots of the characteristic polynomial are

A. Real but not equal **B.** Real and equal
C. Complex conjugates **D.** Imaginary

Q3.94.1 Descriptive 1994 2M

Match List-I with List-II:

List-I

a. Very low response at very high frequencies

b. Overshoot

c. Synchro-control transformer output

List-II

1. Low Pass systems
2. Velocity damping
3. Natural frequency
4. Phase-sensitive modulation
5. Damping ratio

Q3.94.2 Descriptive 1994 2M

The response of an LCR circuit to a step input is:

a. Over damped

b. Critically damped

c. Oscillatory

If the transfer function has:

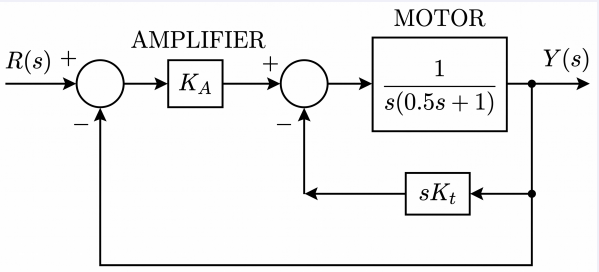
1. poles on the negative real axis
2. poles on the imaginary axis
3. multiple poles on the positive real axis
4. poles on the positive real axis
5. multiple poles on the negative real axis.

Common Data Questions

The next 2 questions share the data below.

Common data for Q3.92.1 and Q3.92.2

Block diagram model of a position control system is shown in fig.



Q3.92.1 Descriptive 1992 8M

In absence of derivative feedback $(K_t = 0)$, determine damping ratio of the system for amplifier gain $K_A = 5$. Also find the steady-state error to unit ramp input.

Q3.92.2 Descriptive 1992 8M

Find suitable values of the parameters K_A and K_t so that damping ratio of the system is increased to 0.7 without affecting the steady-state error as obtained in Q3.92.1.

Q3.91.1 MCQ 1991 2M

A unity-feedback control system has the open-loop transfer function $G(s) = \frac{4(1+2s)}{s^2(s+2)}$. If the input to the system is a unit ramp, the steady-state error will be

A. 0 **B.** 0.5 **C.** 2 **D.** Infinity

Q3.91.2 MCQ 1991 2M

A second-order system has a transfer function given by $G(s) = \dfrac{25}{s^2 + 8s + 25}$. If the system, initially at rest, is subjected to a unit step input at $t = 0$, the second peak in the response will occur at

A. π sec **B.** $\pi/3$ sec **C.** $2\pi/3$ sec **D.** $\pi/2$ sec.

Q3.90.1 MCQ 1990 2M

The steady state error of a stable 'type 0' unity feedback system for a unit step function is

A. 0 **B.** $\dfrac{1}{1 + K_p}$ **C.** ∞ **D.** $\dfrac{1}{K_p}$

Q3.87.1 MCQ 1987 2M

The unity feedback system shown in fig. has:

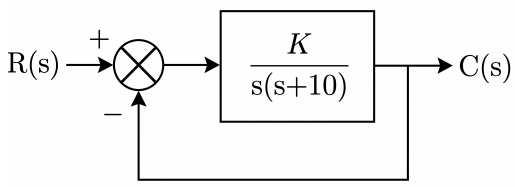


Fig. Q3.87.1

A. Zero steady state position error

B. Zero steady state velocity error

C. Steady state position error $\dfrac{K}{10}$ units

D. Steady state velocity error $\dfrac{K}{10}$ units

Answer Key

Q3.24.1 D	Q3.23.1 A	Q3.22.1 A
Q3.22.2 -0.11 to -0.09	Q3.20.1 30	Q3.19.1 B
Q3.17.1 0.99 to 1.01	Q3.16.1 0.96 to 1.04	Q3.16.2 2.7 to 3.0
Q3.16.3 0.95 to 1.05	Q3.15.1 400	Q3.14.1 38.13 to 38.19
Q3.14.2 C	Q3.14.3 0.49 to 0.51	Q3.13.1 C
Q3.11.1 A	Q3.09.1 B	Q3.08.1 C
Q3.08.2 D	Q3.08.3 C	Q3.07.1 D
Q3.05.1 A	Q3.05.2 C	Q3.04.1 C
Q3.03.1 B	Q3.02.1 C	Q3.02.2 B
Q3.01.1 C	Q3.00.1 *	Q3.00.2 *
Q3.00.3 *	Q3.00.4 *	Q3.00.5 *
Q3.00.6 *	Q3.99.1 B	Q3.99.2 D
Q3.98.1 B	Q3.98.2 A	Q3.98.3 0.2, 0.605, 0.163
Q3.97.1 *	Q3.97.2 *	Q3.97.3 *
Q3.95.1 A	Q3.95.2 C	Q3.94.1 a-1, b-5, c-4
Q3.94.2 a-1, b-5, c-2	Q3.92.1 0.316, 0.2	Q3.92.2 25, 4
Q3.91.1 A	Q3.91.2 A	Q3.90.1 B
Q3.87.1 A		

Note: * descriptive/evidence-based, no single definitive answer.

UNIT

IV

Routh's Stability Criterion

Topics

1. Hurwitz Polynomial and Routh Array
2. Special Cases in Routh Stability
3. Mapping Theorem in Routh-Hurwitz

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 37	**MCQ:** 23	**MSQ:** 1	**NAT:** 12
1M: 8	**2M:** 24	**Descriptive:** 5	**Avg Weightage** ~2M	**Range:** 1M(2020)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT 2026 2M

Consider the unity negative feedback control system shown in the Figure. Value of gain K (> 0) at which the given system will remain marginally stable is

_____

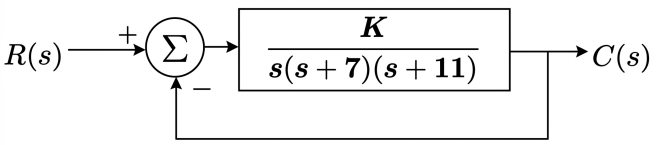


Fig. Q4.26.1

Q4.25.1 MCQ 2025 2M

Consider the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. Let (L, I, R) be defined as follows.

L is the number of roots of $p(s)$ with negative real parts.
I is the number of roots of $p(s)$ that are purely imaginary.
R is the number of roots of $p(s)$ with positive real parts.

Which one of the following options is correct?

A. $L = 2$, $I = 2$, and $R = 1$

B. $L = 3$, $I = 2$, and $R = 0$

C. $L = 1$, $I = 2$, and $R = 2$

D. $L = 0$, $I = 4$, and $R = 1$

Q4.24.1 MCQ 2024 2M

Consider a unity negative feedback control system with forward path gain $G(s) = \dfrac{K}{(s+1)(s+2)(s+3)}$ as shown.

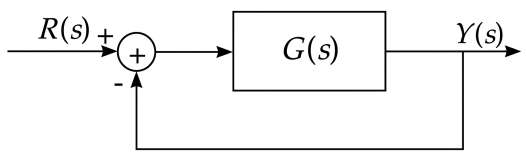


Fig. Q4.24.1

The impulse response of the closed-loop system decays faster than e^{-t} if _____.

A. $1 \le K \le 5$ **B.** $7 \le K \le 21$
C. $-4 \le K \le -1$ **D.** $-24 \le K \le -6$

Q4.22.1 MCQ 2022 2M

Consider an even polynomial $p(s)$ given by

$$p(s) = s^4 + 5s^2 + 4 + K,$$

where K is an unknown real parameter. The complete range of K for which $p(s)$ has all its roots on the imaginary axis is _____.

A. $-4 \le K \le \dfrac{9}{4}$ **B.** $-3 \le K \le \dfrac{9}{2}$
C. $-6 \le K \le \dfrac{5}{4}$ **D.** $-5 \le K \le 0$

Q4.20.1 NAT 2020 1M

The loop transfer function of a negative feedback system is

$$G(s)H(s) = \frac{K(s+11)}{s(s+2)(s+8)}$$

The value of K, for which the system is marginally stable, is

_____

Q4.17.1 NAT 2017 2M

A unity feedback control system is characterized by the open loop transfer function

$$G(s) = \frac{2(s+1)}{s^3 + ks^2 + 2s + 1}$$

The value of k for which the system oscillates at 2 rad/s is

_____

Q4.17.2 MCQ 2017 2M

Which one of the following options correctly describes the locations of the roots of the equation $s^4 + s^2 + 1 = 0$ on the complex plane?

A. Four left half plane (LHP) roots.

B. One right half plane (RHP) root, one LHP root and two roots on the imaginary axis.

C. Two RHP roots and two LHP roots.

D. All four roots are on the imaginary axis.

Q4.16.1 MCQ 2016 1M

Match the inferences X, Y, and Z, about a system, to the corresponding properties of the elements of first column in Routh's Table of the system characteristic equation.

X: The system is stable . . . . . .

Y: The system is unstable . . .

Z: The test breaks down . . . . . .

P: . . . When all elements are positive

Q: . . . When any one element is zero

R: . . . When there is a change in sign of coefficients

A. $X-P, Y-Q, Z-R$ **B.** $X-Q, Y-P, Z-R$
C. $X-R, Y-Q, Z-P$ **D.** $X-P, Y-R, Z-Q$

Q4.16.2 MCQ 2016 2M

The first two rows in the Routh table for the characteristic equation of a certain closed-loop control system are given as

s^3	1	$(2K+3)$
s^2	$2K$	4

The range of K for which the system is stable is

A. $-2.0 < K < 0.5$ **B.** $0 < K < 0.5$
C. $0 < K < \infty$ **D.** $0.5 < K < \infty$

Q4.16.3 NAT 2016 2M

The transfer function of a linear time invariant systems is given by $H(s) = 2s^4 - 5s^3 + 5s - 2$
The number of zeros in the right half of the s-plane is

_____

Q4.15.1 NAT 2015 2M

The characteristic equation of an LTI system is given by $F(s) = s^5 + 2s^4 + 3s^3 + 6s^2 - 4s - 8 = 0$. The number of roots that lie strictly in the left half of s-plane is

_____

Q4.15.2 MCQ 2015 2M

A plant transfer function is given as

$$G(s) = \left(K_p + \frac{K_I}{s}\right)\left(\frac{1}{s(s+2)}\right)$$

When the plant operates in a unity feedback configuration, the condition for the stability of the closed loop system is

A. $K_p > \frac{K_I}{2} > 0$ **B.** $2K_I > K_p > 0$
C. $2K_I < K_p$ **D.** $2K_I > K_p$

Q4.14.1 NAT 2014 1M

The forward path transfer function of a unity negative feedback system is given by $G(s) = \frac{k}{(s+2)(s-1)}$
The value of k which will place both the poles of the closed – loop system at the same location is

_____

Q4.14.2 NAT 2014 2M

Consider a transfer function

$$G_p(s) = \frac{ps^2 + 3ps - 2}{s^2 + (3+p)s + (2-p)}$$

with 'p' a positive real parameter. The maximum value of 'p' until which G_p remains stable is

_____

Q4.13.1 MCQ 2013 1M

A polynomial $f(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x - a_0$ with all coefficients positive has

A. no real roots

B. no negative real root

C. odd number of real roots

D. at least one positive and one negative root

Q4.12.1 MCQ 2012 2M

The feedback system shown below oscillates at 2 rad/sec when

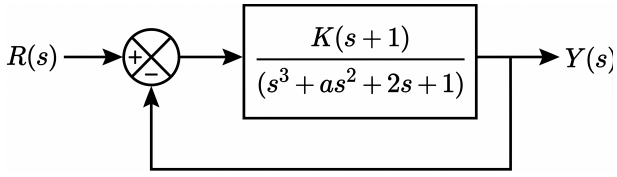


Fig. Q4.12.1

A. $K = 2$ and $a = 0.75$ **B.** $K = 3$ and $a = 0.75$
C. $K = 4$ and $a = 0.5$ **D.** $K = 2$ and $a = 0.5$

Q4.08.1 MCQ 2008 2M

The number of open right half plane poles of $G(s) = \dfrac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3}$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q4.08.2 MCQ 2008 2M

A certain system has transfer function, $G(s) = \dfrac{s+8}{s^2 + \alpha s - 4}$, where α is a parameter. Consider the standard negative unity feedback configuration as shown below.

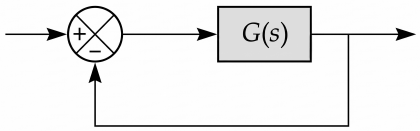


Fig. Q4.08.2

Which of the following statements is true?

A. The closed-loop system is never stable for any value of α.

B. For some positive values of α, the closed-loop system is stable, but not for all positive values.

C. For all positive values of α, the closed-loop system is stable.

D. The closed-loop system is stable for all values of α, both positive and negative.

Q4.07.1 MCQ 2007 1M

If the closed-loop transfer function of a control system is given as, $T(s) = \dfrac{s-5}{(s+2)(s+3)}$. Then it is

A. an unstable system.

B. an uncontrollable system.

C. a minimum phase system.

D. a non-minimum phase system.

Q4.04.1 MCQ 2004 2M

For the polynomial $P(s)$ below,

$$P(s) = s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15,$$

, the number of roots which lie in the right half of the s-plane is

A. 4 **B.** 2 **C.** 3 **D.** 1

Q4.03.1 MCQ 2003 2M

The open-loop transfer function of a unity feedback system is $G(s) = \dfrac{k}{s(s^2 + s + 2)(s + 3)}$, the range of '$k$' for which the system is stable

A. $\dfrac{21}{4} > k > 0$ **B.** $13 > k > 0$
C. $\dfrac{21}{4} < k < \infty$ **D.** $-6 < k < \infty$

Q4.02.1 MCQ 2002 2M

The characteristic polynomial of a system $q(s) = 2s^5 + s^4 + 4s^3 + 2s^2 + 2s + 1$. The system is

A. Stable **B.** Marginally stable
C. Unstable **D.** Oscillatory

Q4.02.2 MCQ 2002 2M

The system shown in the figure remains stable when

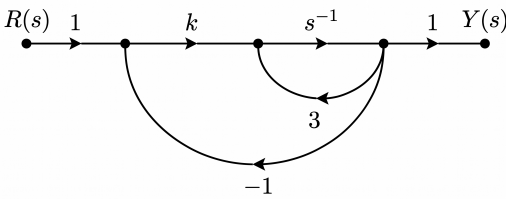


Fig. Q4.02.2

A. $k < -1$ **B.** $-1 < k < 1$
C. $1 < k < 3$ **D.** $k > 3$

Q4.01.1 MCQ 2001 2M

The feedback control system in the figure is stable.

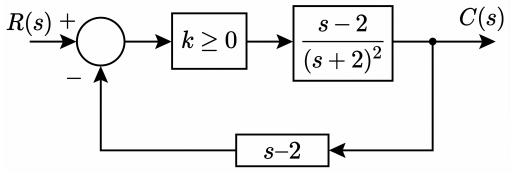


Fig. Q4.01.1

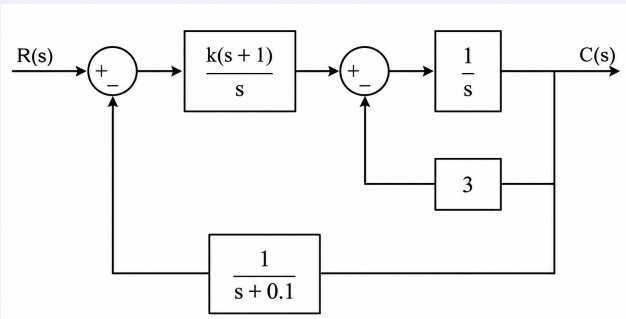
A. for all $k \geq 0$
B. only if $k \geq 1$
C. only if $0 \leq k < 1$
D. only if $0 \leq k \leq 2$

Common Data Questions

The next 2 questions share the data below.

Common data for Q4.01.2 and Q4.01.3

Consider the feedback control system shown in figure



Q4.01.2 NAT 2001 5M

Find the transfer function of the System and its characteristic equation.

Q4.01.3 NAT 2001 5M

Use the Routh-Hurwitz criterion to determine the range of 'K' for which the system is stable.

Q4.00.1 MCQ 2000 1M

A system described by the transfer function

$$H(s) = \frac{1}{s^3 + \alpha s^2 + ks + 3}$$

is stable. The constraints on α and k are,

A. $\alpha > 0, \alpha k < 3$
B. $\alpha > 0, \alpha k > 3$
C. $\alpha < 0, \alpha k > 0$
D. $\alpha > 0, \alpha k < 0$

Q4.00.2 MCQ 2000 1M

An amplifier with resistive negative feedback has two left half plane poles in its open-loop transfer function. The amplifier

A. will always be unstable at high frequency.

B. will be stable for all frequency.

C. may be unstable, depending on the feedback factor.

D. will oscillate at low frequency.

Q4.99.1 NAT 1999 5M

The loop transfer function of a feedback control system is given by

$$G(s)H(s) = \frac{K(s+1)}{s(1+Ts)(1+2s)}, \quad K > 0$$

Using Routh-Hurwitz criterion, determine the region of $K - T$ plane in which the closed-loop system is stable.

Q4.98.1 MCQ 1998 1M

The open loop transfer function of an unity feedback system is $\frac{2s^2 + 6s + 5}{(s+1)^2(s+2)}$. The characteristic equation of the closed loop system is

A. $2s^2 + 6s + 5 = 0$
B. $(s+1)^2(s+2) = 0$
C. $2s^2 + 6s + 5 + (s+1)^2(s+2) = 0$
D. $2s^2 + 6s + 5 - (s+1)^2(s+2) = 0$

Common Data Questions

The next 2 questions share the data below.

Common data for Q4.98.2 and Q4.98.3

The characteristic equation of a feedback control system is $s^4 + 20s^3 + 15s^2 + 2s + K = 0$.

Q4.98.2 NAT 1998 5M

Determine the range of K for the system to be stable.

Q4.98.3 NAT 1998 5M

Can the system be marginally stable? If so, find the required value of K and the frequency of sustained oscillation.

Q4.94.1 Descriptive 1994 2M

If $G(s)$ is a stable transfer function, then $F(s) = \dfrac{1}{G(s)}$ is always a stable transfer function. (T/F).

Q4.93.1 MCQ 1993 2M

If $s^3 + 3s^2 + 4s + A = 0$, then all the roots of this equation are in the left half plane provided that

A. $A > 12$ **B.** $-3 < A < 4$
C. $0 < A < 12$ **D.** $5 < A < 12$.

Q4.90.1 MSQ 1990 2M

An electromechanical closed-loop control system has the following characteristic equation:

$$s^3 + 6Ks^2 + (K + 2)s + 8 = 0$$

where K is the forward gain of the system. The condition for closed-loop stability is

A. $K = 0.528$ **B.** $K = 2$
C. $K = 0$ **D.** $K = -2.258$

Q4.89.1 MCQ 1989 2M

In order to stabilize the system shown in fig. T_i should satisfy:

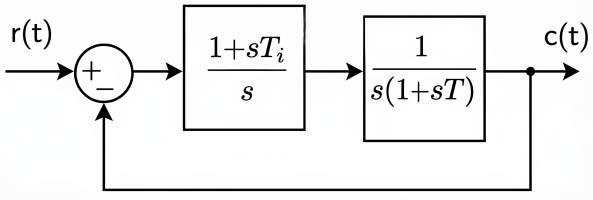


Fig. Q4.89.1

A. $T_i = -T$ **B.** $T_i = T$
C. $T_i < T$ **D.** $T_i > T$

Q4.88.1 MCQ 1988 2M

Consider a characteristic equation given by $s^4 + 3s^3 + 5s^2 + 6s + K + 10 = 0$. The condition for stability is

A. $K > 5$ **B.** $-10 < K$
C. $K > -4$ **D.** $-10 < K < -4$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q4.26.1 1386	Q4.25.1 A	Q4.24.1 A
Q4.22.1 A	Q4.20.1 160	Q4.17.1 0.74 to 0.76
Q4.17.2 C	Q4.16.1 D	Q4.16.2 D
Q4.16.3 3	Q4.15.1 2	Q4.15.2 A
Q4.14.1 2.24 to 2.26	Q4.14.2 1.9 to 2.1	Q4.13.1 D
Q4.12.1 A	Q4.08.1 C	Q4.08.2 C
Q4.07.1 D	Q4.04.1 B	Q4.03.1 A
Q4.02.1 C	Q4.02.2 D	Q4.01.1 C
Q4.01.2 *	Q4.01.3 *	Q4.00.1 B
Q4.00.2 B	Q4.99.1 *	Q4.98.1 C
Q4.98.2 0 to 1.49	Q4.98.3 0.316	Q4.94.1 False
Q4.93.1 C	Q4.90.1 A,B	Q4.89.1 D
Q4.88.1 D		

UNIT

V

Root Locus

Topics

1. Basics of Root Locus
2. Rules for Sketching Root Locus
3. Angle of Departure and Arrival
4. Complementary Root Locus

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 28	**MCQ:** 19	**MSQ:** 1	**NAT:** 8
1M: 8	**2M:** 18	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2025)–2M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q5.25.1 MSQ 2025 1M

Consider the unity-negative-feedback system shown in Figure (i) below, where gain $K \geq 0$. The root locus of this system is shown in Figure (ii) below.
For what value(s) of K will the system in Figure (i) have a pole at $-1 + j1$?

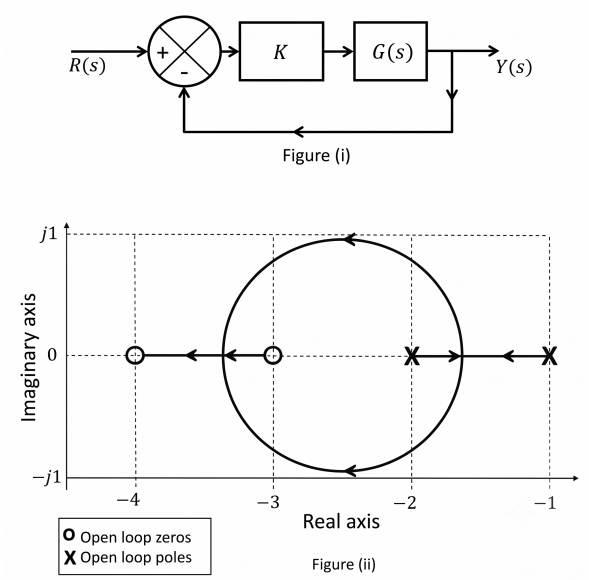


Fig. Q5.25.1

- **A.** $K = 5$
- **B.** $K = 1/5$
- **C.** For no positive value of K
- **D.** For all positive values of K

Q5.22.1 MCQ 2022 1M

The root-locus plot of a closed-loop system with unity negative feedback and transfer function $KG(s)$ in the forward path is shown in the figure. Note that K is varied from 0 to ∞.

Select the transfer function $G(s)$ that results in the root-locus plot of the closed-loop system as shown in the figure.

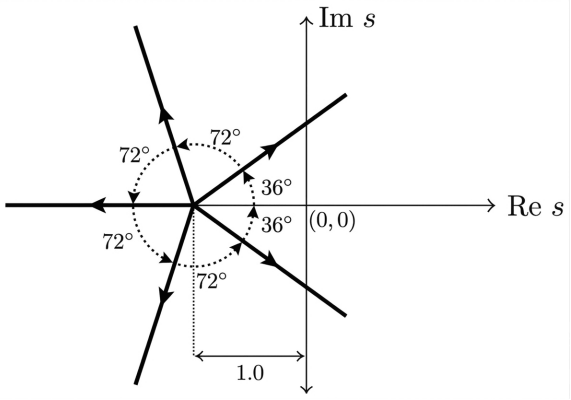


Fig. Q5.22.1

A. $G(s) = \dfrac{1}{(s+1)^5}$ **B.** $G(s) = \dfrac{1}{s^5+1}$

C. $G(s) = \dfrac{s-1}{(s+1)^6}$ **D.** $G(s) = \dfrac{s+1}{s^6+1}$

Q5.20.1 MCQ 2020 2M

The characteristic equation of a system is $s^3 + 3s^2 + (K + 2)s + 3K = 0$ In the root locus plot for the given system, as K varies from 0 to ∞, the break-away or break-in point(s) lie within

A. $(-1, 0)$ **B.** $(-2, -1)$
C. $(-3, -2)$ **D.** $(-\infty, -3)$

Q5.17.1 MCQ 2017 2M

A linear time invariant (LTI) system with the transfer function, $G(s) = \dfrac{K(s^2 + 2s + 2)}{(s^2 - 3s + 2)}$ is connected in unity feedback configuration as shown in the figure.

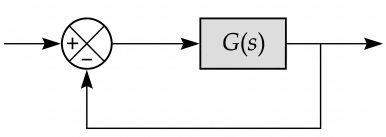


Fig. Q5.17.1

For the closed-loop system shown, the root-locus for $0 < K < \infty$ intersects the imaginary axis for $K = 1.5$. The closed-loop system is stable for

A. $K > 1.5$ **B.** $1 < K < 1.5$
C. $0 < K < 1.5$ **D.** no positive value of K

Q5.16.1 NAT 2016 2M

The forward path transfer function and the feedback path transfer function of a single-loop negative feedback control system are given as, $G(s) = \dfrac{K(s+2)}{s^2+2s+2}$ and $H(s) = 1$ respectively. If the variable parameter K is real positive, then the location of the breakaway point on the root-locus diagram of the system is

---

Q5.16.2 NAT 2016 2M

The open-loop transfer function of a unity-feedback control system is $G(s) = \dfrac{K}{s^2+5s+5}$. The value of K at the breakaway point of the feedback control system's root-locus plot is

---

Q5.15.1 NAT 2015 1M

A unity negative feedback system has the open-loop transfer function $G(s) = \dfrac{K}{s(s+1)(s+3)}$. The value of the gain $K(>0)$ at which the root locus crosses the imaginary axis is

---

Q5.15.2 NAT 2015 2M

For the system shown in the figure, $s = -2.75$ lies on the root locus if K is

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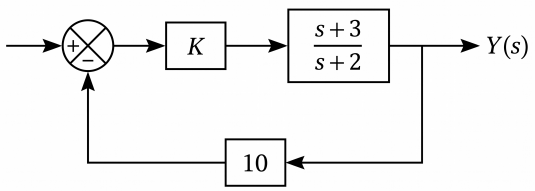

Fig. Q5.15.2

Q5.15.3 NAT 2015 2M

The open-loop transfer function of a unity feedback configuration is given as, $G(s) = \dfrac{K(s+4)}{(s+8)(s^2-9)}$. The value of gain $K\ (>0)$ for which $-1+j2$ lies on the root locus is

---

Q5.14.1 NAT 2014 2M

The characteristic equation of a unity negative feedback system is $1 + KG(s) = 0$. The open-loop transfer function $G(s)$ has one pole at 0 and two poles at -1. The root locus of the system for varying K is shown in the figure.

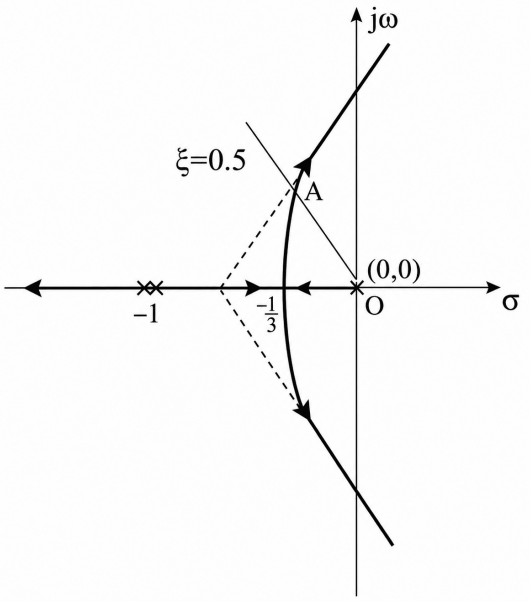

Fig. Q5.14.1

The constant damping ratio line, for $\xi = 0.5$, intersects the root locus at point A. The distance from the origin to point A is given as 0.5. The value of K at point A is

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Q5.14.2 MCQ 2014 2M

In the root locus plot shown in the figure, the pole/zero marks and the arrows have been removed. Which one of the following transfer functions has this root locus?

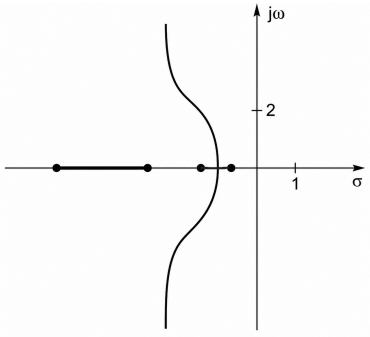


Fig. Q5.14.2

A. $\dfrac{s+1}{(s+2)(s+4)(s+7)}$ **B.** $\dfrac{s+4}{(s+1)(s+2)(s+7)}$

C. $\dfrac{s+7}{(s+1)(s+2)(s+4)}$ **D.** $\dfrac{(s+1)(s+2)}{(s+7)(s+4)}$

Q5.11.1 MCQ 2011 1M

The root locus plot for a system is given below. The open-loop transfer function corresponding to this plot is given by

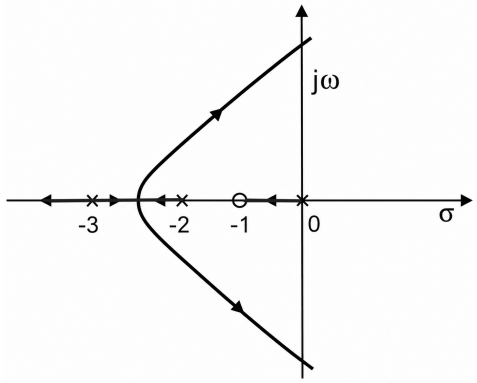


Fig. Q5.11.1

A. $G(s)H(s) = K\dfrac{s(s+1)}{(s+2)(s+3)}$

B. $G(s)H(s) = K\dfrac{(s+1)}{s(s+2)(s+3)^2}$

C. $G(s)H(s) = K\dfrac{1}{s(s-1)(s+2)(s+3)}$

D. $G(s)H(s) = K\dfrac{(s+1)}{s(s+2)(s+3)}$

Q5.09.1 MCQ 2009 2M

The feedback configuration and the pole-zero locations of $G(s) = \dfrac{s^2-2s+2}{s^2+2s+2}$ are shown below.

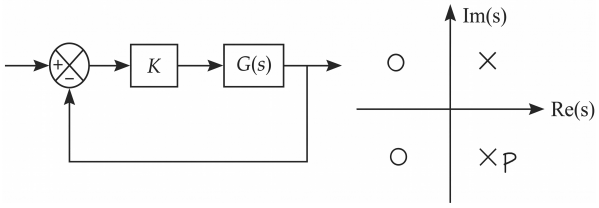


Fig. Q5.09.1

The root locus for negative values of K, i.e. for $-\infty < K <$ 0, has breakaway/break-in points and angle of departure at pole P (with respect to the positive real axis) equal to

A. $\pm\sqrt{2}$ and 0° **B.** $\pm\sqrt{2}$ and 45°

C. $\pm\sqrt{3}$ and 0° **D.** $\pm\sqrt{3}$ and 45°

Q5.07.1 MCQ 2007 2M

A unity feedback control system has an open-loop transfer function

$$G(s) = \frac{K}{s(s^2+7s+12)}$$

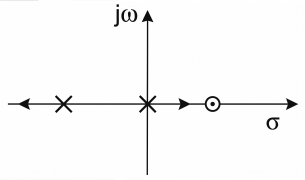
The gain K for which $s = -1 + j1$ will lie on the root locus of this system is

A. 4 **B.** 5.5 **C.** 6.5 **D.** 10

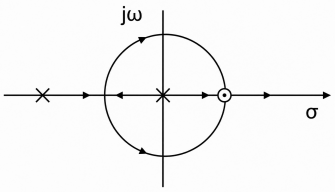
Q5.05.1 MCQ 2005 2M

An unity feedback system is given as, $G(s) = \dfrac{K(1-s)}{s(s+3)}$, indicate the correct root locus diagram.

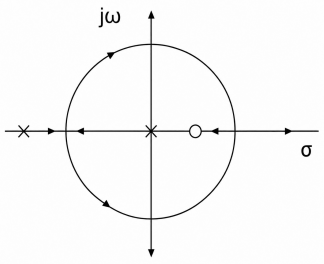
A.



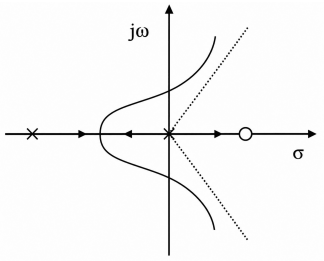
B.



C.



D.



Q5.04.1 MCQ 2004 1M

Given $G(s)H(s) = \dfrac{K}{s(s+1)(s+3)}$, the point of intersection of the asymptotes of the root loci with the real axis is

A. -4 **B.** 1.33 **C.** -1.33 **D.** 4

Q5.03.1 MCQ 2003 2M

The root locus of the system $G(s)H(s) = \dfrac{K}{s(s+2)(s+3)}$, has the break-away point located at

A. $(-0.5, 0)$ **B.** $(-2.548, 0)$
C. $(-4, 0)$ **D.** $(-0.784, 0)$

Q5.02.1 MCQ 2002 1M

Which of the following points is NOT on the root locus of a system with the open-loop transfer function $G(s)H(s) = \dfrac{K}{s(s+1)(s+3)}$?

A. $s = -j\sqrt{3}$ **B.** $s = -1.5$
C. $s = -3$ **D.** $s = -\infty$

Q5.01.1 MCQ 2001 1M

The root-locus diagram for a closed-loop feedback system is shown in figure. The system is overdamped.

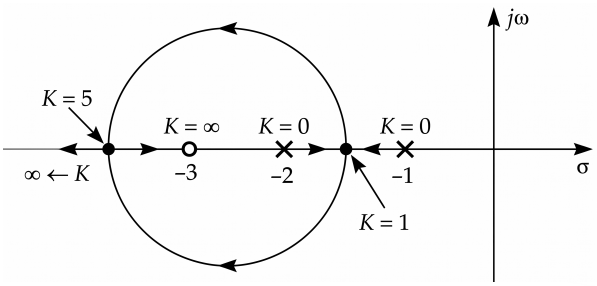


Fig. Q5.01.1

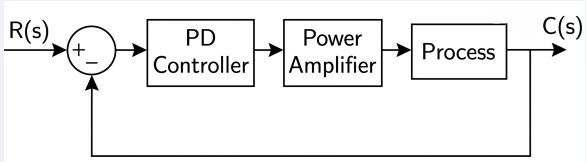
A. only if $0 \le K \le 1$ **B.** only if $1 < K < 5$
C. only if $K > 5$ **D.** if $0 \le K < 1$ or $K > 5$

Common Data Questions

The next 2 questions share the data below.

Common data for questions Q5.01.2 and Q5.01.3

For the feedback control system shown in the figure below, the process transfer function is $G_p(s) = \dfrac{1}{s(s+1)}$, and the amplification factor of the power amplifier is $K \ge 0$. The design specifications required for the system are a time constant of 1 sec and a damping ratio of 0.707.



One question is given in Unit X (Q10.01.1)

Q5.01.2 NAT 2001 5M

Find the desired locations of the closed-loop poles.

Q5.01.3 NAT 2001 5M

Sketch the root locus for the system.

Q5.99.1 MCQ 1999 2M

Consider the points $s_1 = -3 + j4$ and $s_2 = -3 - j2$ in the s-plane. Then, for a system with the open-loop transfer function $G(s)H(s) = \dfrac{K}{(s+1)^4}$

A. s_1 is on the root locus, but not s_2.

B. s_2 is on the root locus, but not s_1.

C. both s_1 and s_2 are on the root locus.

D. neither s_1 nor s_2 is on the root locus.

Q5.94.1 MCQ 1994 1M

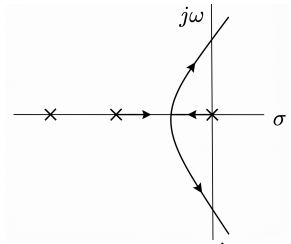
If the open-loop transfer function is a ratio of a numerator polynomial of degree 'm' and a denominator polynomial of degree 'n', then the integer $(n - m)$ represents the number of

A. Breakaway point **B.** Unstable poles
C. Separate root loci **D.** Asymptotes

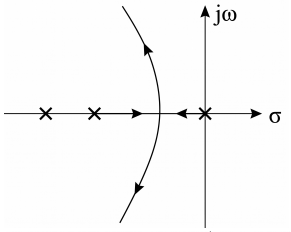
Q5.92.1 MCQ 1992 2M

Given a unity feedback system with open-loop transfer function $G(s) = \dfrac{K}{s(s+1)(s+2)}$. The root locus plot of the system is of the form

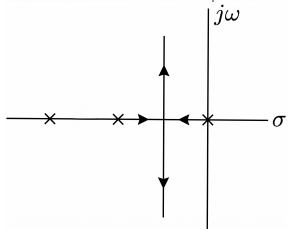
A.



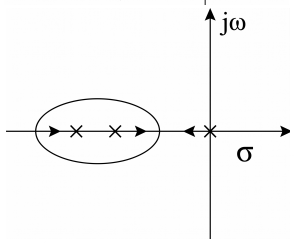
B.



C.



D.



Q5.91.1 MCQ 1991 2M

The characteristic equation of a feedback control system is given by $s^3 + 5s^2 + (K + 6)s + K = 0$, where $K > 0$ is a scalar variable parameter. In the root-locus diagram of the system the asymptotes of the root-loci for large values of K meet at a point in the s-plane, whose coordinates are

A. (-3, 0) **B.** (-2, 0) **C.** (-1, 0) **D.** (2, 0)

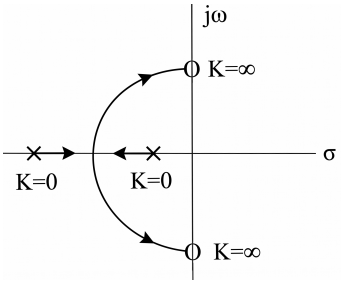
Q5.90.1 MCQ 1990 2M

The transfer function of a closed loop system is

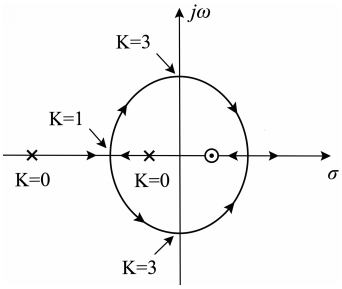
$$T(s) = \frac{K}{s^2 + (3 - K)s + 1}$$

where K is the forward path gain. The root locus plot of the system is

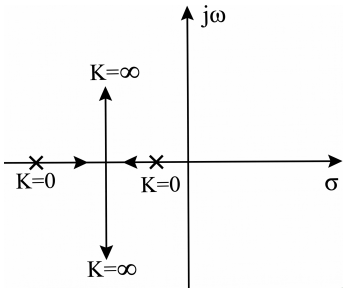
A.



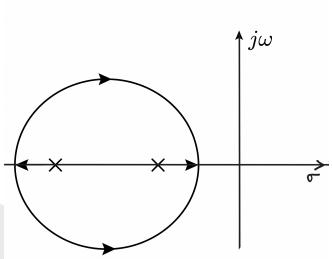
B.



C.



D.



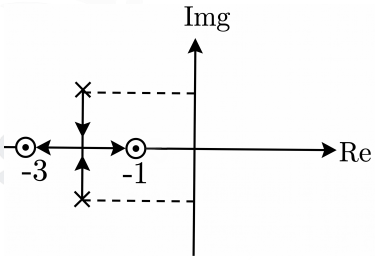
Q5.89.1 MCQ 1989 2M

The OLTF of a feedback system is

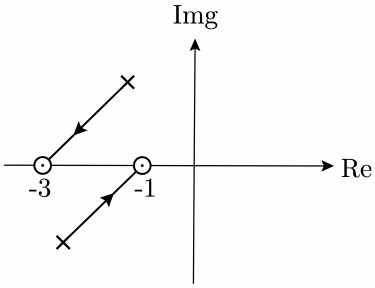
$$G(s)H(s) = \frac{K(s + 1)(s + 3)}{s^2 + 4s + 8}$$

The root locus for the same is

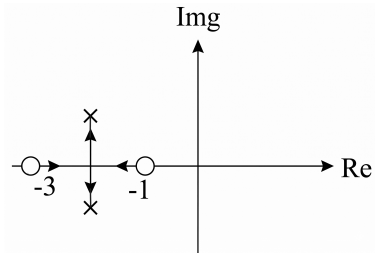
A.



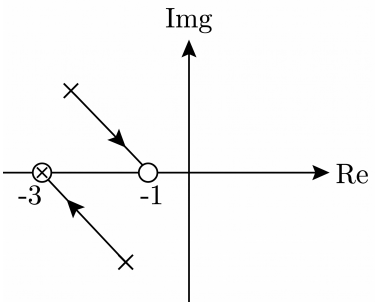
B.



C.



D.



Q5.88.1 MCQ 1988 2M

Consider a closed-loop system shown in fig. (a) below. The root locus for it is shown in fig. (b). The closed loop transfer function for the system is

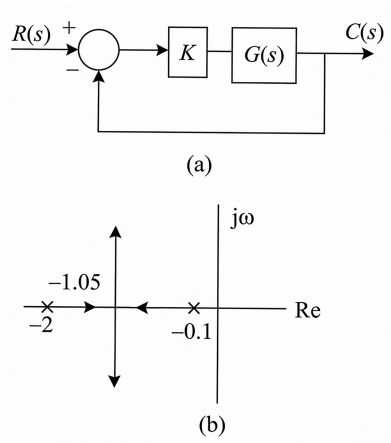


Fig. Q5.88.1

A. $\frac{K}{1+(0.5s+1)(10s+1)}$

B. $\frac{K}{(s+2)(s+0.1)}$

C. $\frac{K}{1+K(0.5s+1)(10s+1)}$

D. $\frac{K}{K+0.2(0.5s+1)(10s+1)}$

Q5.25.1 C	**Q5.22.1** A	**Q5.20.1** A
Q5.17.1 A	**Q5.16.1** -3.45 to -3.35	**Q5.16.2** 1.2 to 1.3 or -1.3 to -1.2
Q5.15.1 12	**Q5.15.2** 0.29 to 0.31	**Q5.15.3** 25 to 26
Q5.14.1 0.32 to 0.41	**Q5.14.2** B	**Q5.11.1** B
Q5.09.1 MTA	**Q5.07.1** D	**Q5.05.1** C
Q5.04.1 C	**Q5.03.1** D	**Q5.02.1** B
Q5.01.1 D	**Q5.01.2** *	**Q5.01.3** *
Q5.99.1 B	**Q5.94.1** D	**Q5.92.1** A
Q5.91.1 B	**Q5.90.1** B	**Q5.89.1** A
Q5.88.1 D		

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

UNIT

VI

Polar Plot & Frequency Response of 2nd Order Systems

Topics

1. Basics of Frequency Response Analysis
2. Frequency Response of 2nd Order Systems
3. Polar Plot
4. Gain and Phase Crossover Frequency
5. Gain Margin and Phase Margin

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 23	**MCQ:** 19	**MSQ:** 0	**NAT:** 4
1M: 9	**2M:** 14	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–2M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q6.23.1 MCQ 2023 1M

The open loop transfer function of a unity negative feedback system is

$$G(s) = \frac{k}{s(1+sT_1)(1+sT_2)},$$

where k, T_1 and T_2 are positive constants. The phase cross-over frequency, in rad/s, is

A. $\frac{1}{\sqrt{T_1T_2}}$ **B.** $\frac{1}{T_1T_2}$ **C.** $\frac{1}{T_1\sqrt{T_2}}$ **D.** $\frac{1}{T_2\sqrt{T_1}}$

Q6.20.1 NAT 2020 2M

A system with transfer function

$$G(s) = \frac{1}{(s+1)(s+a)}$$

where $a > 0$, is subjected to an input $5\cos 3t$. The steady state output of the system is $\frac{1}{\sqrt{10}}\cos(3t - 1.892)$. The value of a is

_____

Q6.18.1 NAT 2018 2M

For a unity feedback control system with the forward path transfer function

$$G(s) = \frac{K}{s(s+2)}$$

The peak resonant magnitude M_r of the closed-loop frequency response is 2. The corresponding value of the gain K (correct to two decimal places) is

_____.

Q6.16.1 NAT 2016 2M

In the feedback system shown below

$$G(s) = \frac{1}{(s+1)(s+2)(s+3)}$$

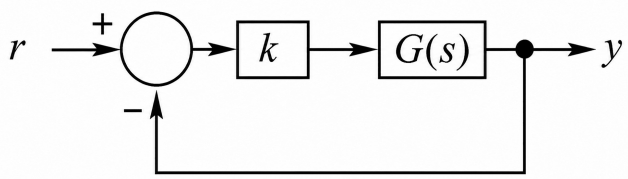


Fig. Q6.16.1

The positive value of k for which the gain margin of the loop is exactly 0 dB and the phase margin of the loop is exactly zero degree is

_____

Q6.15.1 NAT 2015 1M

The phase margin (in degrees) of the system $G(s) = \frac{10}{s(s+10)}$ is

_____.

Q6.15.2 MCQ 2015 1M

The polar plot of the transfer function $G(s) = \frac{10(s+1)}{s+10}$ for $0 \le \omega < \infty$ will be in the

A. first quadrant **B.** second quadrant
C. third quadrant **D.** fourth quadrant

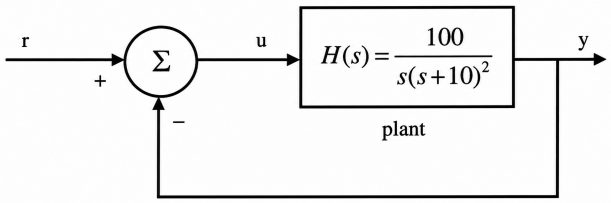
Common Data Questions

The next 1 questions share the data below.

Common Data for Questions Q2.11.1 and Q6.11.1:
The input-output transfer function of a plant is

$$H(s) = \frac{100}{s(s+10)^2}.$$

The plant is placed in a unity negative feedback configuration as shown in the figure below.



The 1st Question is given in Unit II (Q2.11.1)

Q6.11.1 MCQ 2011 2M

The gain margin of the system under closed loop unity negative feedback is

A. 0 dB **B.** 20 dB **C.** 26 dB **D.** 46 dB

Q6.10.1 MCQ 2010 1M

A system with the transfer function $\frac{Y(s)}{X(s)} = \frac{s}{s+p}$ has an output $y(t) = \cos\left(2t - \frac{\pi}{3}\right)$ for the input signal $x(t) = p\cos\left(2t - \frac{\pi}{2}\right)$. Then, the system parameter 'p' is

A. $\sqrt{3}$ **B.** $\frac{2}{\sqrt{3}}$ **C.** 1 **D.** $\frac{\sqrt{3}}{2}$

Q6.08.1 MCQ 2008 2M

The magnitude of frequency response of an underdamped second order system is 5 at 0 rad/sec and peaks to $\frac{10}{\sqrt{3}}$ at $5\sqrt{2}$ rad/sec. The transfer function of the system is

A. $\frac{500}{s^2+10s+100}$ **B.** $\frac{375}{s^2+5s+75}$

C. $\frac{720}{s^2+12s+144}$ **D.** $\frac{1125}{s^2+25s+225}$

Q6.06.1 MCQ 2006 1M

The open-loop transfer function of a unity-gain feedback control system is given by

$$G(s) = \frac{K}{(s+1)(s+2)}.$$

The gain margin of the system in dB is given by

A. 0 **B.** 1 **C.** 20 **D.** ∞

Q6.06.2 MCQ 2006 1M

In the system shown below, $x(t) = (\sin t)\,u(t)$. In steady-state, the response $y(t)$ will be:

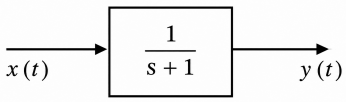


Fig. Q6.06.2

A. $\frac{1}{\sqrt{2}}\sin\left(t-\frac{\pi}{4}\right)$ **B.** $\frac{1}{\sqrt{2}}\sin\left(t+\frac{\pi}{4}\right)$

C. $\frac{1}{\sqrt{2}}e^{-t}\sin t$ **D.** $\sin t - \cos t$

Q6.06.3 MCQ 2006 2M

Consider two transfer functions

$$G_1(s) = \frac{1}{s^2+as+b},$$

and

$$G_2(s) = \frac{s}{s^2+as+b}.$$

The 3-dB bandwidths of their frequency responses, respectively are

A. $\sqrt{a^2-4b}$, $\sqrt{a^2+4b}$ **B.** $\sqrt{a^2+4b}$, $\sqrt{a^2-4b}$

C. $\sqrt{a^2-4b}$, $\sqrt{a^2-4b}$ **D.** $\sqrt{a^2+4b}$, $\sqrt{a^2+4b}$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q6.06.4 and Q6.06.5 using the following data.
Consider a unity-gain feedback control system whose open-loop transfer function is:

$$G(s) = \frac{as+1}{s^2}$$

Q6.06.4 MCQ 2006 2M

The value of "a" so that the system has a phase margin equal to $\frac{\pi}{4}$ is approximately equal to

A. 2.40 **B.** 1.40 **C.** 0.84 **D.** 0.74

Q6.06.5 MCQ 2006 2M

With the value of "a" set for a phase-margin of $\frac{\pi}{4}$, the value of unit-impulse response of the open-loop system at $t = 1$ second is equal to

A. 3.40 **B.** 2.40 **C.** 1.84 **D.** 1.74

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q6.05.1 and Q6.05.2 using the following data.
The open loop transfer function of a unity feedback is given by

$$G(s) = \frac{3e^{-2s}}{s(s+2)}.$$

Q6.05.1 MCQ 2005 2M

The gain and phase crossover frequencies in rad/sec are, respectively

A. 0.632 and 1.26 **B.** 0.632 and 0.485
C. 0.485 and 0.632 **D.** 1.26 and 0.632

Q6.05.2 MCQ 2005 2M

Based on the above results, the gain and phase margins of the system will be

A. −7.09 dB and 87.5° **B.** 7.09 dB and 87.5°
C. 7.09 dB and −87.5° **D.** −7.09 dB and −87.5°

Q6.04.1 MCQ 2004 1M

The gain margin for the system with open-loop transfer function $G(s)\,H(s) = \dfrac{2(1+s)}{s^2}$ is

A. ∞ **B.** 0 **C.** 1 **D.** $-\infty$

Q6.04.2 MCQ 2004 2M

A system has poles at 0.1 Hz, 1 Hz and 80 Hz; zeros at 5 Hz, 100 Hz and 200 Hz. The approximate phase of the system response at 20 Hz is

A. $-90°$ **B.** $0°$ **C.** $90°$ **D.** $-180°$

Q6.03.1 MCQ 2003 2M

The gain margin and the phase margin of feedback system with

$$G(s)\,H(s) = \frac{s}{(s+100)^3}$$

are

A. 0 dB, $0°$ **B.** ∞, ∞
C. ∞, $0°$ **D.** 88.5 dB, ∞

Q6.02.1 MCQ 2002 1M

The phase margin of a system with the open-loop transfer function

$$G(s)\,H(s) = \frac{(1-s)}{(1+s)(2+s)}$$

is

A. $0°$ **B.** $63.4°$ **C.** $90°$ **D.** ∞

Q6.02.2 MCQ 2002 2M

The system with the open loop transfer function

$$G(s)\,H(s) = \frac{1}{s(s^2+s+1)}$$

has a gain margin of

A. -6 dB **B.** 0 dB **C.** 3.5 dB **D.** 6 dB

Q6.94.1 MCQ 1994 1M

The 3-dB bandwidth of a typical second-order system with the transfer function

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

is given by

A. $\omega = \omega_n\sqrt{1-2\zeta^2}$

B. $\omega = \omega_n\sqrt{1-2\zeta^2+\sqrt{\zeta^4-\zeta^2+1}}$

C. $\omega = \omega_n\sqrt{1-2\zeta^2+\sqrt{4\zeta^4-4\zeta^2+2}}$

D. $\omega = \omega_n\sqrt{1-2\zeta^2-\sqrt{4\zeta^4-4\zeta^2+2}}$

Q6.87.1 MCQ 1987 2M

The polar plot of $G(s) = \dfrac{10}{s(s+1)^2}$ intercepts real axis at $\omega = \omega_0$. Then the real part and ω_0 are respectively given by,

A. -2.5, 1 **B.** -5, 0.5 **C.** -5, 1 **D.** -5, 2

Answer Key

Q6.23.1 A	Q6.20.1 3.95 to 4.05	Q6.18.1 14 to 17
Q6.16.1 59.5 to 60.5	Q6.15.1 84 to 84.5	Q6.15.2 A
Q6.11.1 C	Q6.10.1 B	Q6.08.1 A
Q6.06.1 D	Q6.06.2 A	Q6.06.3 C
Q6.06.4 C	Q6.06.5 C	Q6.05.1 D
Q6.05.2 D	Q6.04.1 A	Q6.04.2 A
Q6.03.1 B	Q6.02.1 D	Q6.02.2 B
Q6.94.1 C	Q6.87.1 C	

UNIT

VII

Nyquist Plot

Topics

1. Nyquist Plot
2. Nyquist Stability Criterion
3. Gain Crossover Frequency (ω_{gc}) from Nyquist Plot
4. Phase Crossover Frequency (ω_{pc}) from Nyquist Plot
5. Gain Margin (GM) using Nyquist Plot
6. Phase Margin (PM) using Nyquist Plot

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 20	**MCQ:** 17	**MSQ:** 0	**NAT:** 3
1M: 11	**2M:** 7	**Descriptive:** 2	**Avg Weightage** ~1M	**Range:** 0M(2014)–2M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q7.25.1 MCQ 2025 1M

The Nyquist plot of a system is given in the figure below. Let ω_P, ω_Q, ω_R, and ω_S be the positive frequencies at the points P, Q, R, and S respectively. Which one of the following statements is TRUE?

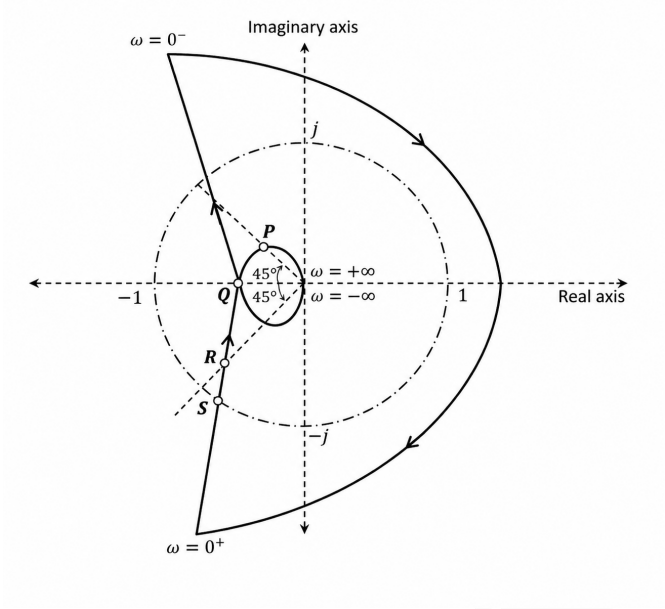


Fig. Q7.25.1

A. ω_S is the gain crossover frequency and ω_P is the phase crossover frequency

B. ω_Q is the gain crossover frequency and ω_R is the phase crossover frequency

C. ω_Q is the gain crossover frequency and ω_S is the phase crossover frequency

D. ω_S is the gain crossover frequency and ω_Q is the phase crossover frequency

Q7.22.1 MCQ 2022 1M

Consider a closed-loop control system with unity negative feedback and $KG(s)$ in the forward path, where the gain $K = 2$. The complete Nyquist plot of the transfer function $G(s)$ is shown in the figure. Note that the Nyquist contour has been chosen to have the clockwise sense. Assume $G(s)$ has no poles on the closed right-half of the complex plane. The number of poles of the closed-loop transfer function in the closed right-half of the complex plane is _____.

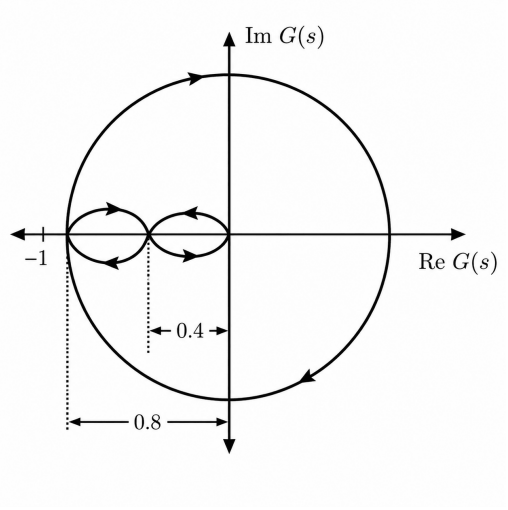


Fig. Q7.22.1

A. 0 **B.** 1 **C.** 2 **D.** 3

Q7.21.1 MCQ 2021 2M

The complete Nyquist plot of the open-loop transfer function $G(s)H(s)$ of a feedback control system is shown in the figure.

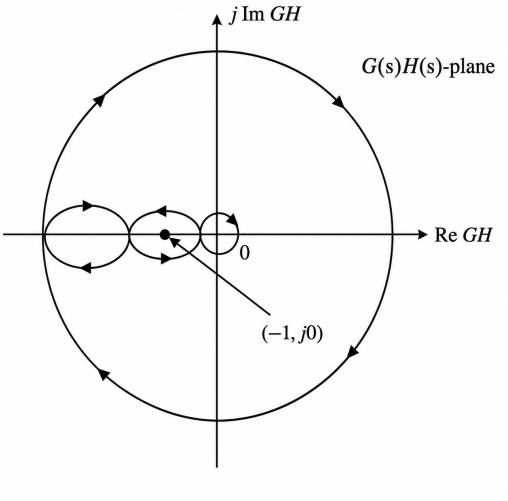


Fig. Q7.21.1

If $G(s)H(s)$ has one zero in the right-half of the s-plane, the number of poles that the closed-loop system will have in the right-half of the s-plane is _____.

A. 0 **B.** 1 **C.** 4 **D.** 3

Q7.20.1 MCQ 2020 1M

The pole-zero map of a rational function $G(s)$ is shown below. When the closed contour Γ is mapped into the $G(s)$-plane, then the mapping encircles _____.

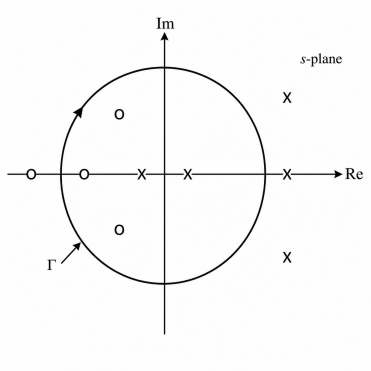


Fig. Q7.20.1

A. the origin of the $G(s)$-plane once in the counter-clockwise direction.

B. the origin of the $G(s)$-plane once in the clockwise direction.

C. the point $-1+j0$ of the $G(s)$-plane once in the counter-clockwise direction.

D. the point $-1+j0$ of the $G(s)$-plane once in the clockwise direction.

Q7.18.1 MCQ 2018 1M

The Nyquist stability criterion and the Routh criterion both are powerful analysis tools for determining the stability of feedback controllers. Identify which of the following statements is FALSE:

A. Both the criteria provide information relative to the stable gain range of the system.

B. The general shape of the Nyquist plot is readily obtained from the Bode magnitude plot for all minimum-phase systems.

C. The Routh criterion is not applicable in the condition of transport lag, which can be readily handled by the Nyquist criterion.

D. The closed-loop frequency response for a unity feedback system cannot be obtained from the Nyquist plot.

Q7.17.1 MCQ 2017 2M

A unity feedback control system is characterized by the open-loop transfer function

$$G(s) = \frac{10K(s+2)}{s^3 + 3s^2 + 10}$$

The Nyquist path and the corresponding Nyquist plot of $G(s)$ are shown in the figures below.

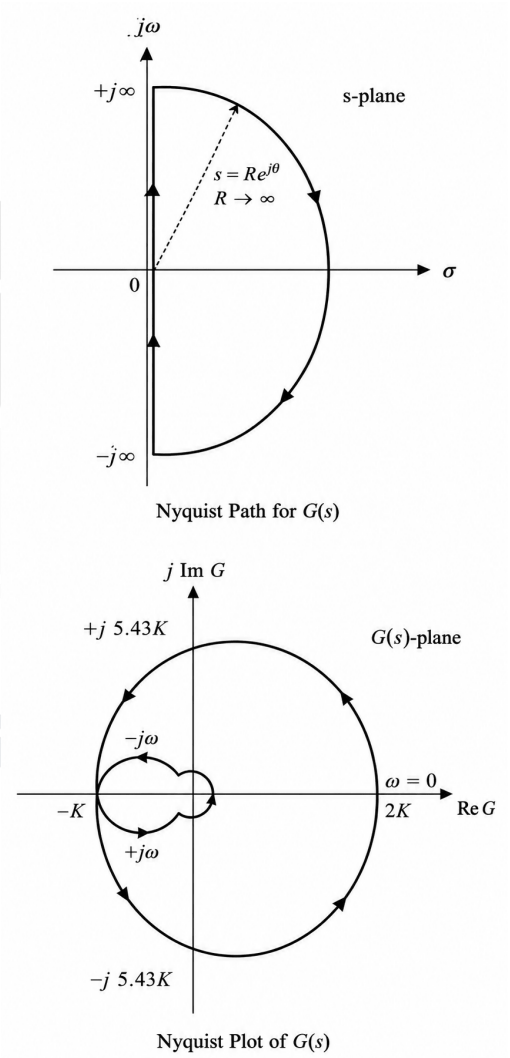


Fig. Q7.17.1

If $0 < K < 1$, then the number of poles of the closed-loop transfer function that lie in the right-half of the s-plane is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q7.17.2 MCQ 2017 2M

The Nyquist plot of the transfer function

$$G(s) = \frac{K}{(s^2 + 2s + 2)(s + 2)}$$

does not encircle the point $(-1 + j0)$ for $K = 10$ but does encircle the point $(-1 + j0)$ for $K = 100$. Then the closed loop system (having unity gain feedback) is

A. stable for $K = 10$ and stable for $K = 100$

B. stable for $K = 10$ and unstable for $K = 100$

C. unstable for $K = 10$ and stable for $K = 100$

D. unstable for $K = 10$ and unstable for $K = 100$

Q7.16.1 MCQ 2016 1M

The number and direction of encirclements around the point $-1 + j0$ in the complex plane by the Nyquist plot of

$$G(s) = \frac{1 - s}{4 + 2s}$$

is

A. zero
B. one, anti-clockwise
C. one, clockwise
D. two, clockwise

Q7.16.2 MCQ 2016 1M

A closed-loop control system is stable if the Nyquist plot of the corresponding open-loop transfer function

A. encircles the s-plane point $(-1 + j0)$ in the counter-clockwise direction as many times as the number of right-half s-plane poles.

B. encircles the s-plane point $(0 - j1)$ in the clockwise direction as many times as the number of right-half s-plane poles.

C. encircles the s-plane point $(-1 + j0)$ in the counter-clockwise direction as many times as the number of left-half s-plane poles.

D. encircles the s-plane point $(-1 + j0)$ in the counter-clockwise direction as many times as the number of right-half s-plane zeros.

Q7.14.1 MCQ 2014 1M

Consider the feedback system shown in the figure. The Nyquist plot of $G(s)$ is also shown. Which one of the following conclusions is correct?

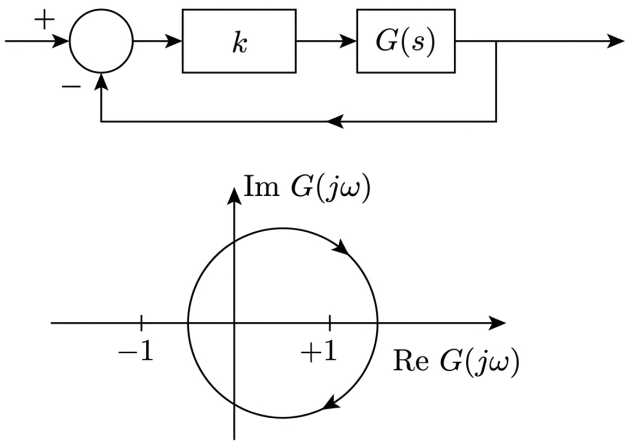


Fig. Q7.14.1

A. $G(s)$ is an all-pass filter

B. $G(s)$ is a strictly proper transfer function

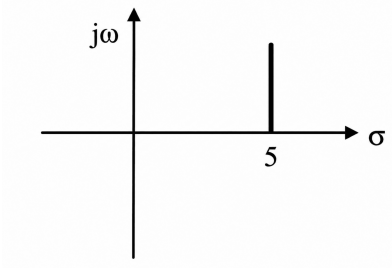
C. $G(s)$ is a stable and minimum-phase transfer function

D. The closed-loop system is unstable for sufficiently large and positive k

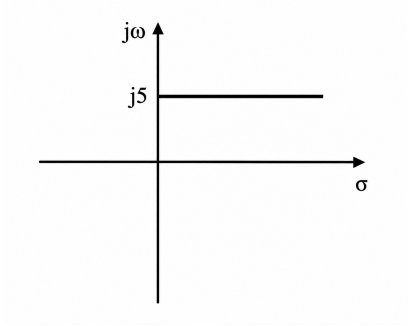
Q7.11.1 MCQ 2011 1M

For the transfer function $G(j\omega) = 5 + j\omega$, the corresponding Nyquist plot for positive frequency has the form

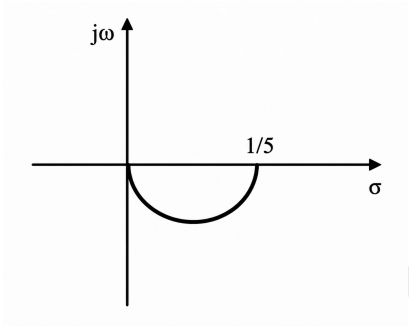
A.



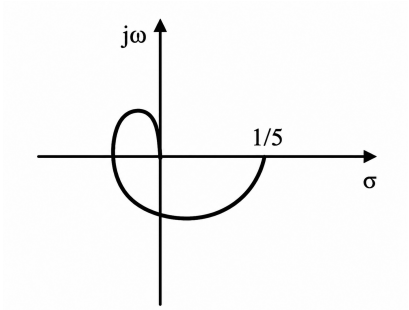
B.



C.



D.

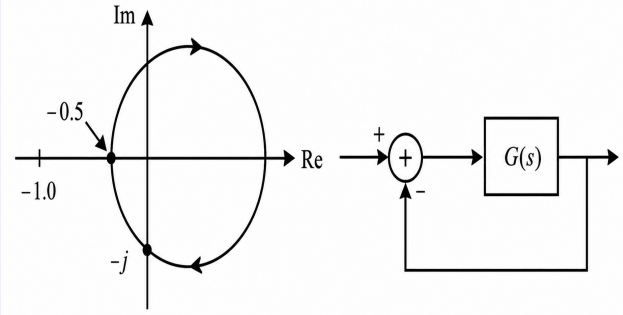


Common Data Questions

The next 2 questions share the data below.

Answer Questions Q7.09.1 and Q7.09.2 using the following data.

The Nyquist plot of a stable transfer function $G(s)$ is shown in the figure. We are interested in the stability of the closed loop system in the feedback configuration shown.



Q7.09.1 MCQ 2009 2M

Which of the following statements is true?

A. $G(s)$ is an all-pass filter

B. $G(s)$ has a zero in the right-half plane

C. $G(s)$ is the impedance of a passive network

D. $G(s)$ is marginally stable

Q7.09.2 MCQ 2009 2M

The gain and phase margins of $G(s)$ for closed loop stability are

A. 6 dB and 180°
B. 3 dB and 180°
C. 6 dB and 90°
D. 3 dB and 90°

Q7.06.1 MCQ 2006 2M

The Nyquist plot of $G(j\omega)H(j\omega)$ for a closed-loop control system, passes through $(-1, j0)$ point in the GH plane. The gain margin of the system in dB is equal to

A. infinite
B. greater than zero
C. less than zero
D. zero

Q7.05.1 MCQ 2005 2M

The polar diagram of a conditionally stable system for open loop gain $K = 1$ is shown in figure. The open loop transfer function of the system is known to be stable. The closed loop system is stable for

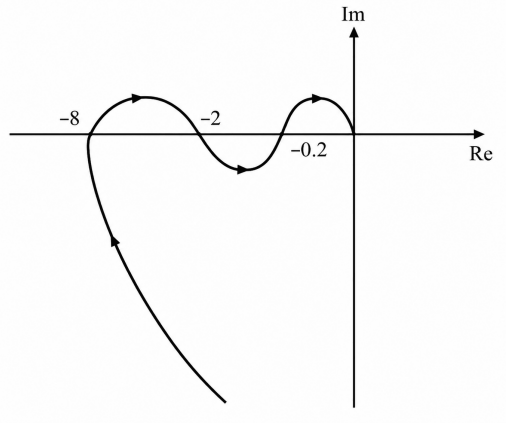


Fig. Q7.05.1

A. $K < 5$ and $\frac{1}{2} < K < \frac{1}{8}$

B. $K < \frac{1}{8}$ and $\frac{1}{2} < K < 5$

C. $K < \frac{1}{8}$ and $5 < K$

D. $K > \frac{1}{8}$ and $K < 5$

Q7.03.1 MCQ 2003 1M

Figure shows the Nyquist plot of the open-loop transfer function $G(s)H(s)$ of a system. If $G(s)H(s)$ has one right hand pole, the closed loop system is

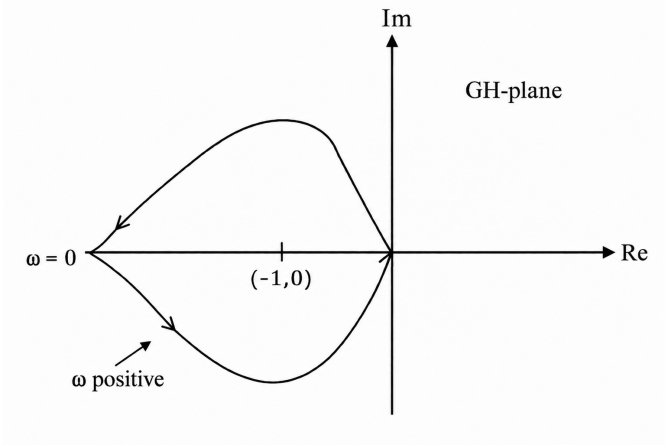


Fig. Q7.03.1

A. always stable

B. unstable with one closed loop right hand pole

C. unstable with two closed loop right hand poles

D. unstable with three closed loop right hand poles

Q7.02.1 NAT 2002 5M

The Nyquist plot of an all-pole second order open-loop system is shown in the below figure. Obtain the transfer function of the system.

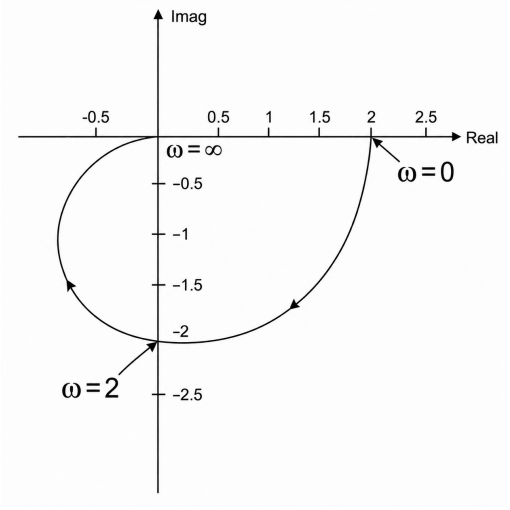


Fig. Q7.02.1

Q7.01.1 MCQ 2001 1M

The Nyquist plot for the open-loop transfer function $G(s)$ of a unity negative feedback system is shown in below figure. If $G(s)$ has no pole in the right half of s-plane, the number of roots of the system characteristic equation in the right half of s-plane is

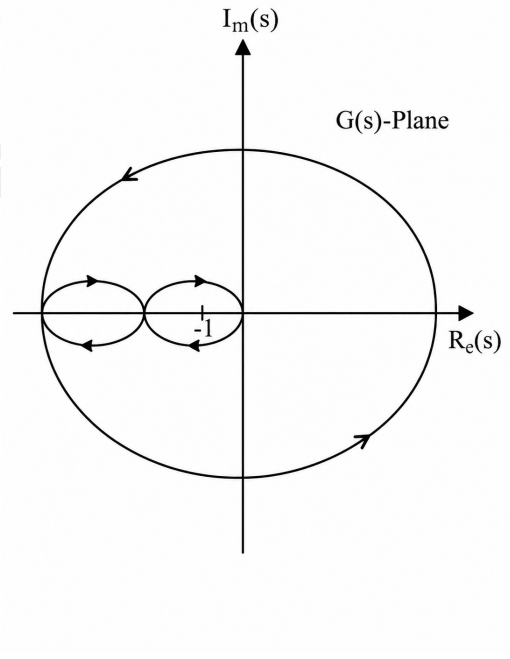


Fig. Q7.01.1

A. 0 **B.** 1 **C.** 2 **D.** 3

Q7.94.1 NAT 1994 1M

The open loop frequency response of a system at two particular frequencies are given by:

$$1.2\angle -180^\circ \quad \text{and} \quad 1.0\angle -190^\circ$$

The closed loop unity feedback control system is then (Stable/Unstable). Enter 0 for unstable and 1 for stable.

_____.

Q7.92.1 NAT 1992 5M

A unity feedback system has open loop transfer function

$$G(s) = \frac{1}{s(2s+1)(s+1)}$$

Sketch Nyquist plot for the system and there from obtain the gain margin and the phase margin.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q7.25.1 D	Q7.22.1 C	Q7.21.1 D
Q7.20.1 B	Q7.18.1 D	Q7.17.1 C
Q7.17.2 B	Q7.16.1 A	Q7.16.2 A
Q7.14.1 D	Q7.11.1 A	Q7.09.1 B
Q7.09.2 C	Q7.06.1 D	Q7.05.1 B
Q7.03.1 A	Q7.02.1 *	Q7.01.1 A
Q7.94.1 0	Q7.92.1 *	

UNIT

VIII

Bode Plot

Topics

1. Bode Gain and Phase Plots
2. Minimum and Non-Minimum Phase Systems
3. Bode Plots for Repeated Poles and Zeros
4. Recovery of Transfer Function from Bode Plot
5. Error in Asymptotic Bode Gain Plot
6. Bode Plots of 2nd order systems

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 21	**MCQ:** 13	**MSQ:** 0	**NAT:** 8
1M: 9	**2M:** 11	**Descriptive:** 1	**Avg Weightage** ~2M	**Range:** 1M(2024)–2M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q8.24.1 MCQ 2024 1M

In the context of Bode magnitude plots, 40 dB/decade is the same as _____.

A. 12 dB/octave
B. 6 dB/octave
C. 20 dB/octave
D. 10 dB/octave

Q8.23.1 NAT 2023 2M

The asymptotic magnitude Bode plot of a minimum phase system is shown in the figure. The transfer function of the system is $G(s) = \dfrac{k(s+z)^a}{s^b(s+p)^c}$, where k, z, p, a, b and c are positive constants. The value of $(a+b+c)$ is

_____(rounded off to the nearest integer).

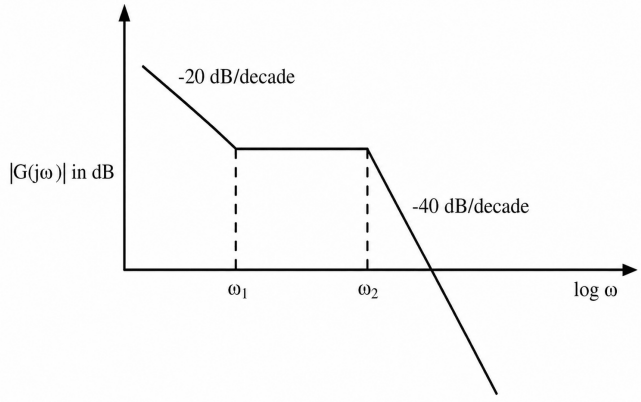


Fig. Q8.23.1

Q8.19.1 MCQ 2019 1M

For an LTI system, the Bode plot for its gain is as illustrated in the figure shown. The number of system poles N_p and the number of system zeros N_z in the frequency range $1\text{ Hz} \le f \le 10^7\text{ Hz}$ is

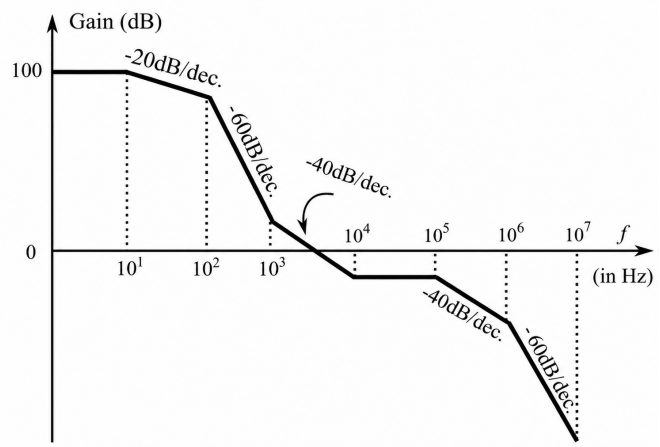


Fig. Q8.19.1

A. $N_p = 5,\ N_z = 2$
B. $N_p = 6,\ N_z = 3$
C. $N_p = 7,\ N_z = 4$
D. $N_p = 4,\ N_z = 2$

Q8.18.1 NAT 2018 2M

The figure below shows the Bode magnitude and phase plots of a stable transfer function

$$G(s) = \frac{n_0}{s^3 + d_2 s^2 + d_1 s + d_0}.$$

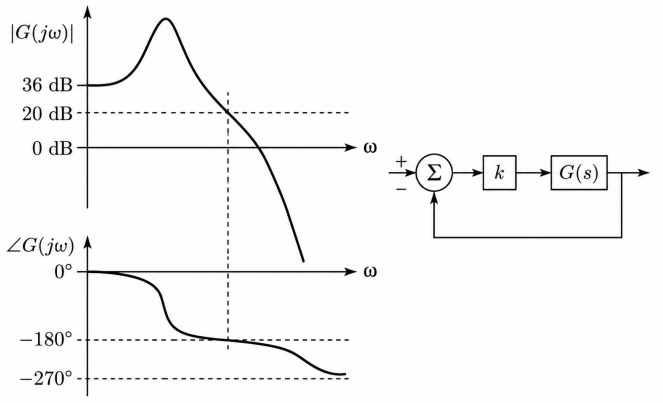


Fig. Q8.18.1

Consider the negative unity feedback configuration with gain k in the feedforward path. The closed loop is stable for $k < k_0$. The maximum value of k_0 is

_____.

Q8.17.1 MCQ 2017 1M

Consider a stable system with transfer function

$$G(s) = \frac{s^p + b_1 s^{p-1} + \cdots + b_p}{s^q + a_1 s^{q-1} + \cdots + a_q}$$

where $b_1, \ldots, b_p$ and $a_1, \ldots, a_q$ are real valued constants. The slope of the Bode log magnitude curve of $G(s)$ converges to -60 dB/decade as $\omega \to \infty$. A possible pair of values for p and q is

A. $p = 0$ and $q = 3$
B. $p = 1$ and $q = 7$
C. $p = 2$ and $q = 3$
D. $p = 3$ and $q = 5$

Q8.16.1 NAT 2016 2M

The asymptotic Bode phase plot of

$$G(s) = \frac{k}{(s + 0.1)(s + 10)(s + p_1)}$$

with k and p_1 both positive, is shown below.

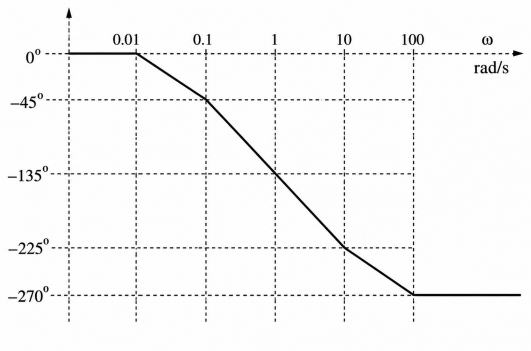


Fig. Q8.16.1

The value of p_1 is

_____

Q8.15.1 NAT 2015 2M

The transfer function of a mass-spring-damper system is given by

$$G(s) = \frac{1}{Ms^2 + Bs + K}$$

The frequency response data for the system are given in the following table.

ω in rad/s	$\lvert G(j\omega)\rvert$ in dB	$\arg(G(j\omega))$ in deg
0.01	−18.5	−0.2
0.1	−18.5	−1.3
0.2	−18.4	−2.6
1	−16	−16.9
2	−11.4	−89.4
3	−21.5	−151
5	−32.8	−167
10	−45.3	−174.5

The unit step response of the system approaches a steady state value of

_____.

Q8.15.2 NAT 2015 1M

Consider the Bode plot shown in the figure. Assume that all the poles and zeros are real-valued.

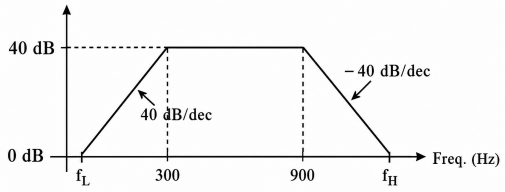


Fig. Q8.15.2

The value of $f_H - f_L$ (in Hz) is

_____.

Q8.14.1 MCQ 2014 1M

In a Bode magnitude plot, which one of the following slopes would be exhibited at high frequencies by a 4^{th} order all-pole system?

A. −80 dB/decade
B. −40 dB/decade
C. +40 dB/decade
D. +80 dB/decade

Q8.14.2 NAT 2014 2M

The Bode asymptotic magnitude plot of a minimum phase system is shown in the figure.

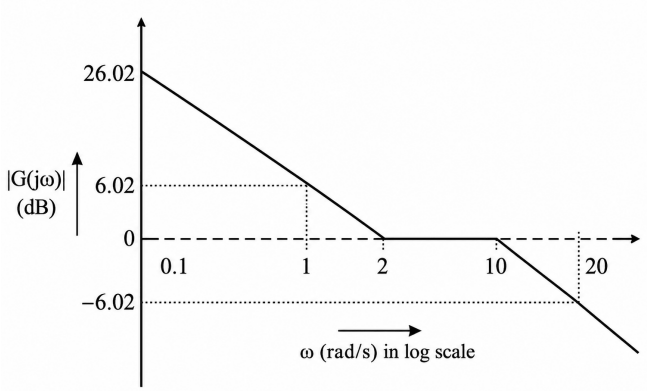


Fig. Q8.14.2

If the system is connected in a unity negative feedback configuration, the steady state error of the closed loop system, to a unit ramp input, is

_____.

Q8.14.3 NAT 2014 2M

The phase margin in degrees of

$$G(s) = \frac{10}{(s+0.1)(s+1)(s+10)}$$

calculated using the asymptotic Bode plot is

_____.

Q8.13.1 MCQ 2013 1M

The Bode plot of a transfer function $G(s)$ is shown in the figure below.

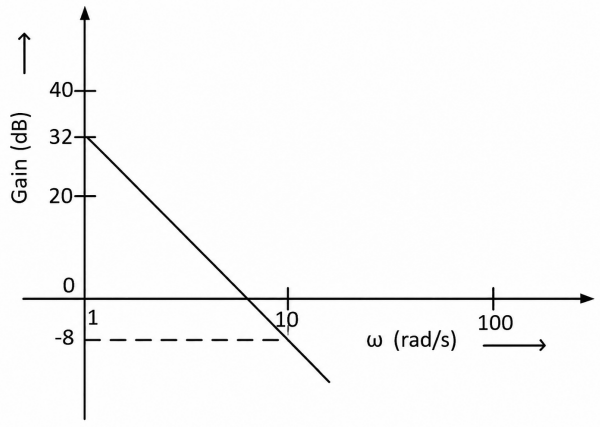


Fig. Q8.13.1

The gain $(20\log|G(s)|)$ is 32 dB and −8 dB at 1 rad/s and 10 rad/s respectively. The phase is negative for all ω. Then $G(s)$ is

A. $\frac{39.8}{s}$ **B.** $\frac{39.8}{s^2}$ **C.** $\frac{32}{s}$ **D.** $\frac{32}{s^2}$

Q8.10.1 MCQ 2010 1M

For the asymptotic Bode magnitude plot shown below, the system transfer function can be

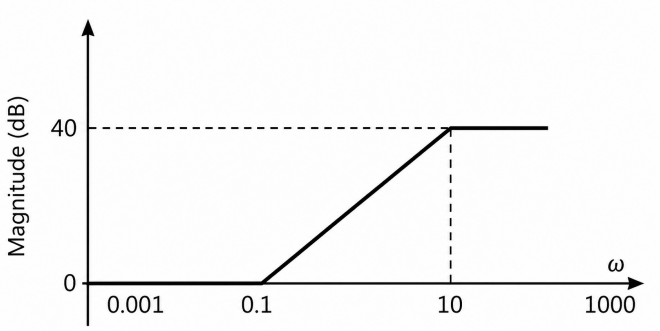


Fig. Q8.10.1

A. $\frac{10s+1}{0.1s+1}$ **B.** $\frac{100s+1}{0.1s+1}$ **C.** $\frac{100s}{10s+1}$ **D.** $\frac{0.1s+1}{10s+1}$

Q8.07.1 MCQ 2007 2M

The asymptotic Bode plot of a transfer function is as shown in the figure. The transfer function $G(s)$ corresponding to this Bode plot is

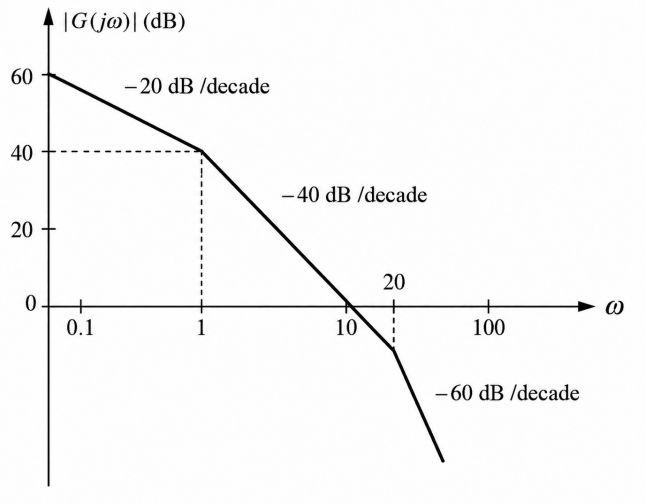


Fig. Q8.07.1

A. $\frac{1}{(s+1)(s+20)}$ **B.** $\frac{1}{s(s+1)(s+20)}$

C. $\frac{100}{s(s+1)(s+20)}$ **D.** $\frac{100}{s(s+1)(1+0.05s)}$

Q8.04.1 MCQ 2004 2M

Consider the Bode magnitude plot shown in the figure. The transfer function $H(s)$ is

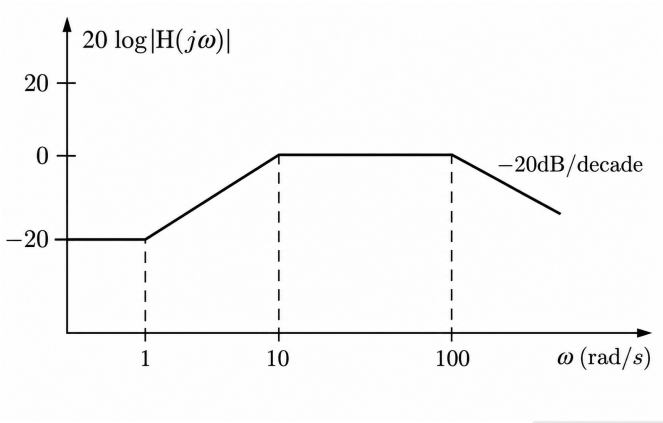


Fig. Q8.04.1

A. $\dfrac{(s+10)}{(s+1)(s+100)}$ **B.** $\dfrac{10(s+10)}{(s+1)(s+100)}$

C. $\dfrac{10^2(s+1)}{(s+10)(s+100)}$ **D.** $\dfrac{10^3(s+100)}{(s+1)(s+10)}$

Q8.03.1 MCQ 2003 2M

The approximate Bode magnitude plot of a minimum-phase system is shown in figure. The transfer function of the system is

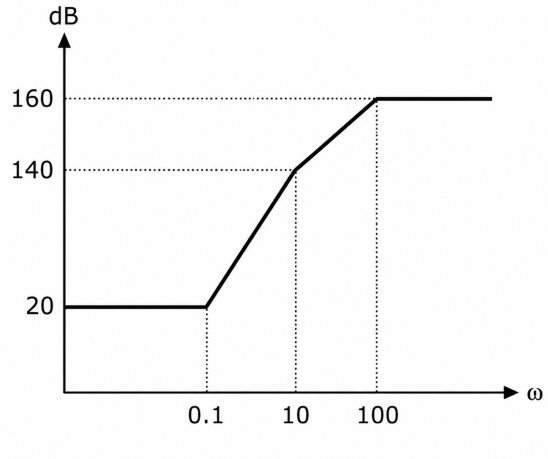


Fig. Q8.03.1

A. $10^8\dfrac{(s+0.1)^3}{(s+10)^2(s+100)}$ **B.** $10^7\dfrac{(s+0.1)^3}{(s+10)(s+100)}$

C. $10^8\dfrac{(s+0.1)^2}{(s+10)^2(s+100)}$ **D.** $10^9\dfrac{(s+0.1)^3}{(s+10)(s+100)^2}$

Q8.99.1 NAT 1999 5M

The asymptotic Bode plot of the minimum phase open-loop transfer function $G(s)H(s)$ is as shown in the below figure. Obtain the transfer function $G(s)H(s)$.

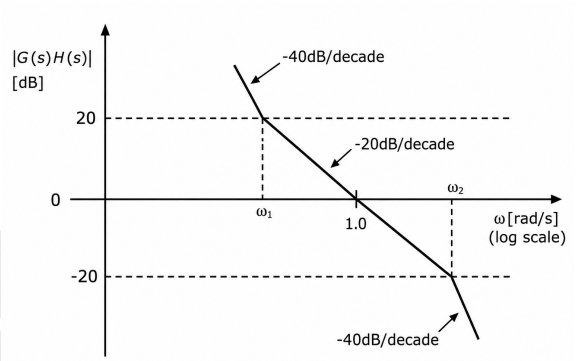


Fig. Q8.99.1

Q8.98.1 MCQ 1998 1M

In the Bode plot of a unity feedback control system, the value of phase of $G(j\omega)$ at the gain crossover frequency is $-125°$. The phase margin of the system is

A. $-125°$ **B.** $-55°$ **C.** $55°$ **D.** $125°$

Q8.95.1 MCQ 1995 1M

Non-minimum phase transfer function is defined as the transfer function

A. which has zeros in the right half s-plane.

B. which has zeros only in the left half s-plane.

C. which has poles in the right half s-plane.

D. which has poles in the left half s-plane.

Q8.92.1 MCQ 1992 2M

Bode plot of a stable system is shown in figure. The transfer function of the system is

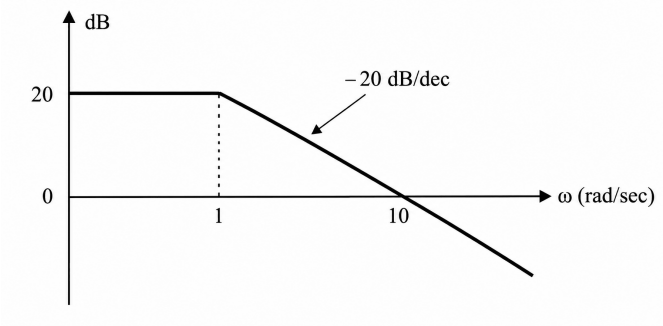


Fig. Q8.92.1

A. $\frac{10}{1+s}$ **B.** $\frac{100}{1+s}$ **C.** $\frac{10s}{1+s}$ **D.** $\frac{100s}{1+s}$

Q8.87.1 MCQ 1987 2M

A system has fourteen poles and two zeros. Its high frequency asymptote in its magnitude plot, has a slope of

A. −40 dB/dec **B.** −240 dB/dec
C. −280 dB/dec **D.** −320 dB/dec

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q8.24.1 A	Q8.23.1 4 to 4	Q8.19.1 B
Q8.18.1 0.1 to 0.1	Q8.17.1 A	Q8.16.1 0.95 to 1.05
Q8.15.1 0.10 to 0.13	Q8.15.2 8970 to 8970	Q8.14.1 A
Q8.14.2 0.49 to 0.51	Q8.14.3 42 to 48	Q8.13.1 B
Q8.10.1 A	Q8.07.1 D	Q8.04.1 C
Q8.03.1 A	Q8.99.1 *	Q8.98.1 C
Q8.95.1 A	Q8.92.1 A	Q8.87.1 B

UNIT

IX

State Space Analysis

Topics

1. State Variables and State Diagram
2. State Space Representation
3. Eigen Values and Eigen Vectors in State Space
4. Solution of State Space Equations
5. State Transition Matrix and Its Properties
6. Recovery of Transfer Function from State Space
7. Similarity Transforms
8. Controllable and Observable Canonical Forms
9. Parallel and Cascade Decomposition
10. Controllability — Kalman's and Gilbert's Test
11. Observability — Kalman's and Gilbert's Test
12. Observable and Detectable Systems

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 32	**MSQ:** 4	**NAT:** 9
1M: 5	**2M:** 32	**Descriptive:** 8	**Avg Weightage** ~2M	**Range:** 1M(2016)–4M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q9.26.1 MCQ 2026 2M

The state and output equations for a control system are:

$$\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} x$$

Which of the following expressions correctly represents the transfer function $\frac{Y(s)}{U(s)}$ of the system with zero initial conditions?

A. $\frac{3s}{s^2 + 4s - 6}$ **B.** $\frac{3s + 5}{s^2 + 4s - 6}$

C. $\frac{3s + 5}{s^2 + 4s + 6}$ **D.** $\frac{3s}{s^2 + 4s + 6}$

Q9.25.1 MSQ 2025 2M

Consider a system where $x_1(t)$, $x_2(t)$, and $x_3(t)$ are three internal state signals and $u(t)$ is the input signal. The differential equations governing the system are given by

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t).$$

Which of the following statements is/are TRUE?

☐ **A.** The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are bounded for all bounded inputs

☐ **B.** There exists a bounded input such that at least one of the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ is unbounded

☐ **C.** There exists a bounded input such that the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded

☐ **D.** The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded for all bounded inputs

Q9.24.1 MSQ 2024 2M

Consider a system S represented in state space as

$$\frac{dx}{dt} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r, \quad y = \begin{bmatrix} 2 & -5 \end{bmatrix} x.$$

Which of the state space representations given below has/have the same transfer function as that of S?

☐ **A.** $\frac{dx}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r, \; y = \begin{bmatrix} 1 & 2 \end{bmatrix} x$

☐ **B.** $\frac{dx}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r, \; y = \begin{bmatrix} 0 & 2 \end{bmatrix} x$

☐ **C.** $\frac{dx}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} -1 \\ 3 \end{bmatrix} r, \; y = \begin{bmatrix} 1 & 1 \end{bmatrix} x$

☐ **D.** $\frac{dx}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} r, \; y = \begin{bmatrix} 1 & 2 \end{bmatrix} x$

Q9.23.1 MCQ 2023 2M

The state equation of a second order system is

$$\dot{\mathbf{x}}(t) = A\mathbf{x}(t), \quad \mathbf{x}(0) \text{ is the initial condition.}$$

Suppose λ_1 and λ_2 are two distinct eigenvalues of A and $\mathbf{v}_1$ and $\mathbf{v}_2$ are the corresponding eigenvectors. For constants α_1 and α_2, the solution, $\mathbf{x}(t)$, of the state equation is

A. $\sum_{i=1}^{2} \alpha_i e^{\lambda_i t}\mathbf{v}_i$ **B.** $\sum_{i=1}^{2} \alpha_i e^{2\lambda_i t}\mathbf{v}_i$

C. $\sum_{i=1}^{2} \alpha_i e^{3\lambda_i t}\mathbf{v}_i$ **D.** $\sum_{i=1}^{2} \alpha_i e^{4\lambda_i t}\mathbf{v}_i$

Q9.20.1 MCQ 2020 2M

For the given circuit, which one of the following is the correct state equation?

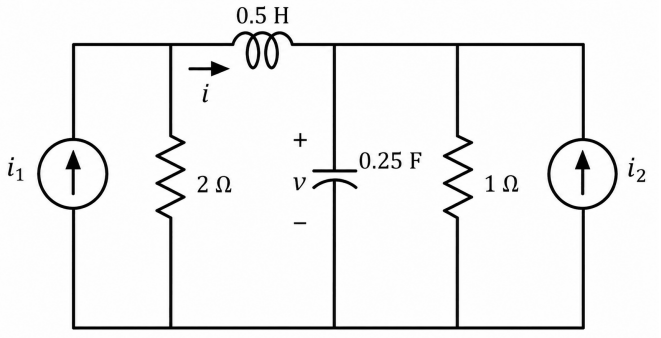


Fig. Q9.20.1

A. $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

B. $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

C. $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

D. $\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$

Q9.19.1 MCQ 2019 2M

Let the state-space representation of an LTI system be $\dot{\mathbf{x}}(t) = A\,\mathbf{x}(t) + B\,u(t)$, $y(t) = C\,\mathbf{x}(t) + d\,u(t)$ where A, B, C are matrices, d is a scalar, $u(t)$ is the input to the system, and $y(t)$ is its output. Let $B = \begin{bmatrix}0 & 0 & 1\end{bmatrix}^T$ and $d = 0$. Which one of the following options for A and C will ensure that the transfer function of this LTI system is

$$H(s) = \frac{1}{s^3 + 3s^2 + 2s + 1} \quad ?$$

A. $A = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -2 & -3\end{bmatrix}$ and $C = \begin{bmatrix}1 & 0 & 0\end{bmatrix}$

B. $A = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -3 & -2 & -1\end{bmatrix}$ and $C = \begin{bmatrix}1 & 0 & 0\end{bmatrix}$

C. $A = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -2 & -3\end{bmatrix}$ and $C = \begin{bmatrix}0 & 0 & 1\end{bmatrix}$

D. $A = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -3 & -2 & -1\end{bmatrix}$ and $C = \begin{bmatrix}0 & 0 & 1\end{bmatrix}$

Q9.17.1 NAT 2017 1M

Consider the state space realization

$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix} = \begin{bmatrix}0 & 0\\ 0 & -9\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}0\\ 45\end{bmatrix}u(t),$$

with the initial condition

$$\begin{bmatrix}x_1(0)\\ x_2(0)\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix},$$

where $u(t)$ denotes the unit step function. The value of $\lim_{t\to\infty}\sqrt{x_1^2(t) + x_2^2(t)}$ is

_____.

Q9.17.2 MCQ 2017 2M

A second-order LTI system is described by the following state equations,

$$\frac{d}{dt}x_1(t) - x_2(t) = 0$$

$$\frac{d}{dt}x_2(t) + 2x_1(t) + 3x_2(t) = r(t)$$

where $x_1(t)$ and $x_2(t)$ are the two state variables and $r(t)$ denotes the input. The output $c(t) = x_1(t)$. The system is

A. undamped (oscillatory) **B.** underdamped
C. critically damped **D.** overdamped

Q9.16.1 MCQ 2016 2M

A second-order linear time-invariant system is described by the following state equations

$$\frac{d}{dt}x_1(t) + 2x_1(t) = 3u(t)$$

$$\frac{d}{dt}x_2(t) + x_2(t) = u(t)$$

where $x_1(t)$ and $x_2(t)$ are the two state variables and $u(t)$ denotes the input. If the output $c(t) = x_1(t)$, then the system is

A. controllable but not observable

B. observable but not controllable

C. both controllable and observable

D. neither controllable nor observable

Q9.15.1 MCQ 2015 2M

The state variable representation of a system is given as

$$\dot{x} = \begin{bmatrix}0 & 1\\ 0 & -1\end{bmatrix}x; \qquad x(0) = \begin{bmatrix}1\\ 0\end{bmatrix}$$

$$y = \begin{bmatrix}0 & 1\end{bmatrix}x$$

The response $y(t)$ is

A. $\sin(t)$ **B.** $1 - e^t$ **C.** $1 - \cos(t)$ **D.** 0

Q9.14.1 MCQ 2014 2M

The state transition matrix $\phi(t)$ of a system

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

is

A. $\begin{bmatrix}t & 1\\ 1 & 0\end{bmatrix}$ **B.** $\begin{bmatrix}1 & 0\\ t & 1\end{bmatrix}$ **C.** $\begin{bmatrix}0 & 1\\ 1 & t\end{bmatrix}$ **D.** $\begin{bmatrix}1 & t\\ 0 & 1\end{bmatrix}$

Q9.14.2 MCQ 2014 2M

The state equation of a second-order linear system is given by

$$\dot{x}(t) = Ax(t), \qquad x(0) = x_0$$

For

$$x_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$$

and for

$$x_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad x(t) = \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix}$$

When

$$x_0 = \begin{bmatrix} 3 \\ 5 \end{bmatrix},$$

$x(t)$ is

A. $\begin{bmatrix} -8e^{-t} + 11e^{-2t} \\ 8e^{-t} - 22e^{-2t} \end{bmatrix}$ **B.** $\begin{bmatrix} 11e^{-t} - 8e^{-2t} \\ -11e^{-t} + 16e^{-2t} \end{bmatrix}$

C. $\begin{bmatrix} 3e^{-t} - 5e^{-2t} \\ -3e^{-t} + 10e^{-2t} \end{bmatrix}$ **D.** $\begin{bmatrix} 5e^{-t} - 3e^{-2t} \\ -5e^{-t} + 6e^{-2t} \end{bmatrix}$

Q9.14.3 MCQ 2014 2M

An unforced linear time invariant (LTI) system is represented by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

If the initial conditions are $x_1(0) = 1$ and $x_2(0) = -1$, the solution of the state equation is

A. $x_1(t) = -1,\ x_2(t) = 2$

B. $x_1(t) = -e^{-t},\ x_2(t) = 2e^{-t}$

C. $x_1(t) = e^{-t},\ x_2(t) = -e^{-2t}$

D. $x_1(t) = -e^{-t},\ x_2(t) = -2e^{-t}$

Q9.14.4 MCQ 2014 2M

Consider the state space system expressed by the signal flow diagram shown in the figure.

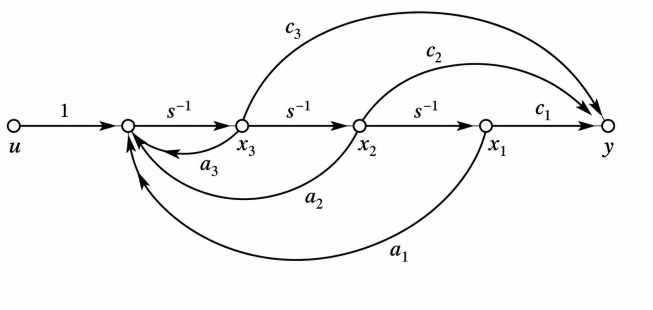


Fig. Q9.14.4

The corresponding system is

A. always controllable **B.** always observable
C. always stable **D.** always unstable

Q9.14.5 MCQ 2014 2M

Consider the state space model of a system, as given below

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The system is

A. controllable and observable

B. uncontrollable and observable

C. uncontrollable and unobservable

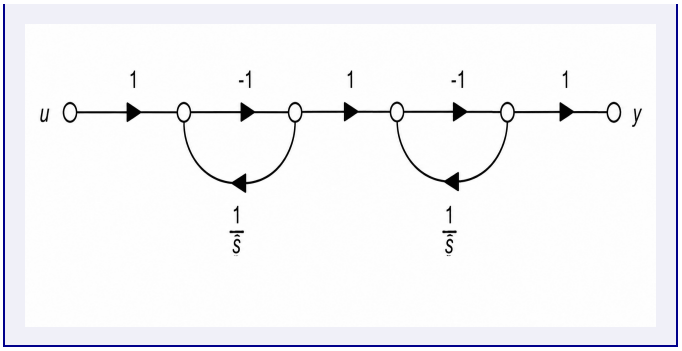
D. controllable and unobservable

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.13.1 and Q9.13.2 using the following data.
The state diagram of a system is shown below. A system is described by the state-variable equations

$$\dot{\mathbf{X}} = \mathbf{AX} + \mathbf{B}u; \qquad y = \mathbf{CX} + \mathbf{D}u$$



Q9.13.1 MCQ 2013 2M

The state-variable equations of the system shown in the figure above are

A. $\dot{\mathbf{X}} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \mathbf{X} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = \begin{bmatrix} 1 & -1 \end{bmatrix} \mathbf{X} + u$

B. $\dot{\mathbf{X}} = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix} \mathbf{X} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = \begin{bmatrix} -1 & -1 \end{bmatrix} \mathbf{X} + u$

C. $\dot{\mathbf{X}} = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix} \mathbf{X} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = \begin{bmatrix} -1 & -1 \end{bmatrix} \mathbf{X} + u$

D. $\dot{\mathbf{X}} = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix} \mathbf{X} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$

$y = \begin{bmatrix} 1 & -1 \end{bmatrix} \mathbf{X} + u$

Q9.13.2 MCQ 2013 2M

The state transition matrix e^{At} of the system shown in the figure above is

A. $\begin{bmatrix} e^{-t} & 0 \\ te^{-t} & e^{-t} \end{bmatrix}$ **B.** $\begin{bmatrix} e^{-t} & 0 \\ -te^{-t} & e^{-t} \end{bmatrix}$

C. $\begin{bmatrix} e^{-t} & 0 \\ e^{-t} & e^{-t} \end{bmatrix}$ **D.** $\begin{bmatrix} e^{-t} & -te^{-t} \\ 0 & e^{-t} \end{bmatrix}$

Q9.12.1 MCQ 2012 2M

The state variable description of an LTI system is given by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

where y is the output and u is the input. The system is controllable for

A. $a_1 \neq 0,\ a_2 = 0,\ a_3 \neq 0$ **B.** $a_1 = 0,\ a_2 \neq 0,\ a_3 \neq 0$

C. $a_1 = 0,\ a_2 \neq 0,\ a_3 = 0$ **D.** $a_1 \neq 0,\ a_2 \neq 0,\ a_3 = 0$

Q9.11.1 MCQ 2011 2M

The block diagram of a system with one input u and two outputs y_1 and y_2 is given below.

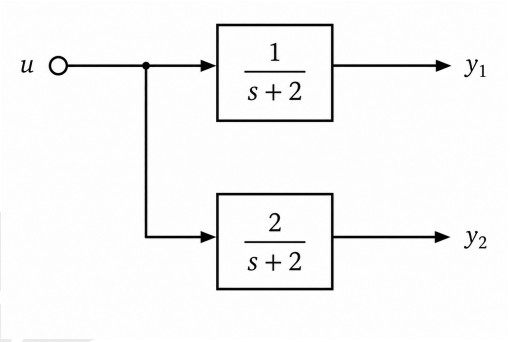


Fig. Q9.11.1

A state space model of the above system in terms of the state vector x and the output vector

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}^T$$

is

A. $\dot{x} = [2]x + [1]u;$

$y = [1 \quad 2]x$

B. $\dot{x} = [-2]x + [1]u;$

$y = \begin{bmatrix} 1 \\ 2 \end{bmatrix} x$

C. $\dot{x} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u;$

$y = [1 \quad 2]x$

D. $\dot{x} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u;$

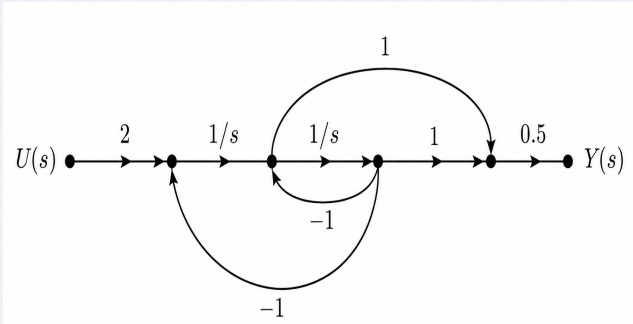
$y = \begin{bmatrix} 1 \\ 2 \end{bmatrix} x$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.10.1 and Q9.10.2 using the following data.

The signal flow graph of a system is shown below.



Q9.10.1 MCQ 2010 2M

The state variable representation of the system can be

A. $\dot{x} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$

$y = \begin{bmatrix} 0 & 0.5 \end{bmatrix} x$

B. $\dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$

$y = \begin{bmatrix} 0 & 0.5 \end{bmatrix} x$

C. $\dot{x} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$

$y = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} x$

D. $\dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$

$y = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} x$

Q9.10.2 MCQ 2010 2M

The transfer function of the system is

A. $\frac{s+1}{s^2+1}$ **B.** $\frac{s-1}{s^2+1}$ **C.** $\frac{s+1}{s^2+s+1}$ **D.** $\frac{s-1}{s^2+s+1}$

Q9.09.1 MCQ 2009 1M

Consider the system

$$\frac{dx}{dt} = Ax + Bu$$

with

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} p \\ q \end{bmatrix}$$

where p and q are arbitrary real numbers. Which of the following statements about the controllability of the system is true?

A. The system is completely state controllable for any nonzero values of p and q

B. Only $p = 0$ and $q = 0$ result in controllability

C. The system is uncontrollable for all values of p and q

D. We cannot conclude about controllability from the given data

Q9.08.1 MCQ 2008 2M

A signal flow graph of a system is given below.

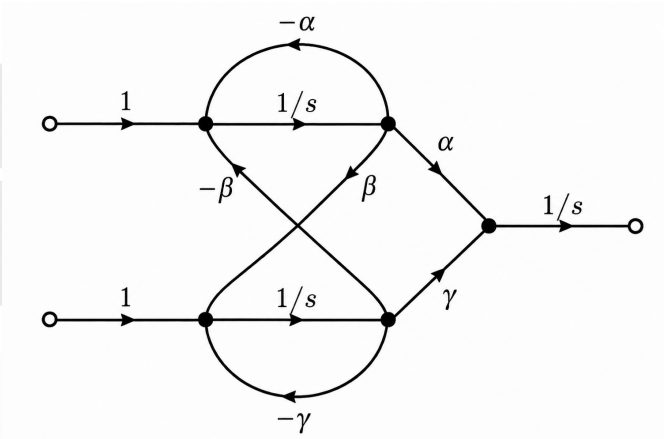


Fig. Q9.08.1

The set of equations that correspond to this signal flow graph is

A. $\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \beta & -\gamma & 0 \\ \gamma & \alpha & 0 \\ -\alpha & -\beta & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

B. $\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & \alpha & \gamma \\ 0 & -\alpha & -\gamma \\ 0 & \beta & -\beta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

C. $\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\alpha & -\beta & 0 \\ \beta & -\gamma & 0 \\ \alpha & \gamma & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

D. $\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\gamma & 0 & \beta \\ \gamma & 0 & \alpha \\ -\beta & 0 & -\alpha \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

Q9.07.1 MCQ 2007 2M

The state space representation of a separately excited DC servo motor dynamics is given as

$$\begin{bmatrix} \frac{d\omega}{dt} \\ \frac{di_a}{dt} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & -10 \end{bmatrix} \begin{bmatrix} \omega \\ i_a \end{bmatrix} + \begin{bmatrix} 0 \\ 10 \end{bmatrix} u$$

where ω is the speed of the motor, i_a is the armature current and u is the armature voltage. The transfer function $\frac{\omega(s)}{U(s)}$ of the motor is

A. $\frac{10}{s^2 + 11s + 11}$ **B.** $\frac{1}{s^2 + 11s + 11}$

C. $\frac{10s + 10}{s^2 + 11s + 11}$ **D.** $\frac{1}{s^2 + s + 1}$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.07.2 and Q9.07.3 using the following data.

Consider a linear system whose state space representation is

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t).$$

If the initial state vector of the system is

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix},$$

then the system response is

$$\mathbf{x}(t) = \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix}.$$

If the initial state vector of the system changes to

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

then the system response becomes

$$\mathbf{x}(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}.$$

Q9.07.2 MCQ 2007 2M

The eigenvalue and eigenvector pairs (λ_i, v_i) for the system are

A. $\left(-1, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$ and $\left(-2, \begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)$

B. $\left(-2, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$ and $\left(-1, \begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)$

C. $\left(-1, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$ and $\left(2, \begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)$

D. $\left(-2, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$ and $\left(1, \begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)$

Q9.07.3 MCQ 2007 2M

The system matrix **A** is

A. $\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$

C. $\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$ **D.** $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

Q9.06.1 MCQ 2006 2M

A linear system is described by the following state equation

$$\dot{X}(t) = AX(t) + BU(t), \qquad A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

The state-transition matrix of the system is:

A. $\begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$ **B.** $\begin{bmatrix} -\cos t & \sin t \\ -\sin t & -\cos t \end{bmatrix}$

C. $\begin{bmatrix} -\cos t & -\sin t \\ -\sin t & \cos t \end{bmatrix}$ **D.** $\begin{bmatrix} \cos t & -\sin t \\ \cos t & \sin t \end{bmatrix}$

Q9.05.1 MSQ 2005 1M

A linear system is equivalently represented by two sets of state equations -

$$\dot{X} = AX + BU \quad \text{and} \quad \dot{W} = CW + DU$$

The eigen values of the representations are also compared as $[\lambda]$ and $[\mu]$. Which one of the following statements is true?

A. $[\lambda] = [\mu]$ and $X = W$ **B.** $[\lambda] = [\mu]$ and $X \neq W$

C. $[\lambda] \neq [\mu]$ and $X = W$ **D.** $[\lambda] \neq [\mu]$ and $X \neq W$

Q9.04.1 MCQ 2004 2M

The state variable equations of a system are:

$$1.\ \dot{x}_1 = -3x_1 - x_2 + u$$
$$2.\ \dot{x}_2 = 2x_1$$
$$y = x_1 + u$$

the system is

A. controllable but not observable

B. observable but not controllable

C. neither controllable nor observable

D. controllable and observable

Q9.04.2 MCQ 2004 2M

Given

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

the state transition matrix e^{At} is given by

A. $\begin{bmatrix} 0 & e^{-t} \\ e^{-t} & 0 \end{bmatrix}$ **B.** $\begin{bmatrix} e^{t} & 0 \\ 0 & e^{t} \end{bmatrix}$

C. $\begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-t} \end{bmatrix}$ **D.** $\begin{bmatrix} 0 & e^{t} \\ e^{t} & 0 \end{bmatrix}$

Q9.03.1 MCQ 2003 2M

The zero-input response of a system given by the state-space equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

is

A. $\begin{bmatrix} te^{t} \\ t \end{bmatrix}$ **B.** $\begin{bmatrix} e^{t} \\ t \end{bmatrix}$ **C.** $\begin{bmatrix} e^{t} \\ te^{t} \end{bmatrix}$ **D.** $\begin{bmatrix} t \\ te^{t} \end{bmatrix}$

Q9.02.1 MCQ 2002 1M

The transfer function $\frac{Y(s)}{U(s)}$ of a system described by the state equations

$$\dot{x}(t) = -2x(t) + 2u(t)$$
$$y(t) = 0.5x(t)$$

is

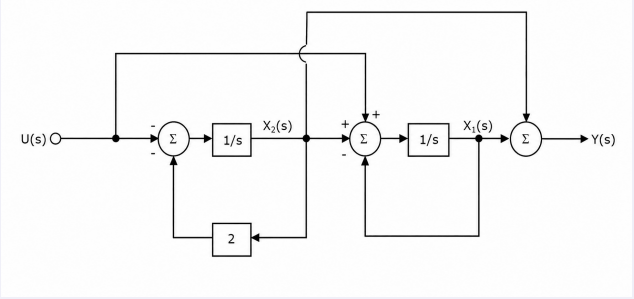
A. $\frac{0.5}{(s-2)}$ **B.** $\frac{1}{(s-2)}$ **C.** $\frac{0.5}{(s+2)}$ **D.** $\frac{1}{(s+2)}$

Common Data Questions

The next 3 questions share the data below.

Answer Questions Q9.02.2, Q9.02.3 and Q9.02.4 using the following data. These three questions together carry a total of **5 marks**.

The block diagram of a linear time invariant system is given in the below figure.



Q9.02.2 NAT 2002

Write down the state variable equations for the system in matrix form assuming the state vector to be $[x_1(t)\ x_2(t)]^T$.

Q9.02.3 NAT 2002

Find out the state transition matrix.

Q9.02.4 NAT 2002

Determine $y(t)$, $t \geq 0$, when the initial values of the state at time $t = 0$ are $x_1(0) = 1$, and $x_2(0) = 1$.

Q9.01.1 MCQ 2001 2M

For the system described by the state equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0.5 & 1 & 2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

If the control signal u is given by

$$u = \begin{bmatrix} -0.5 & -3 & -5 \end{bmatrix} x + v$$

then the eigen values of the closed loop system will be

A. 0, −1, −2 **B.** 0, −1, −3

C. −1, −1, −2 **D.** 0, −1, −1

Common Data Questions

The next 3 questions share the data below.

Answer Questions Q9.00.1, Q9.00.2 and Q9.00.3 using the following data. These three questions together carry a total of **5 marks**.

A certain linear, time-invariant system has the state and output representation shown below:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-2 & 1\\ 0 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 0\end{bmatrix} u$$

$$y = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Q9.00.1 NAT 2000

Find the eigenvalues (natural frequencies) of the system.

Q9.00.2 NAT 2000

If $u(t) = \delta(t)$ and $x_1(0^+) = x_2(0^+) = 0$, find $x_1(t)$, $x_2(t)$ and $y(t)$, for $t > 0$.

Q9.00.3 NAT 2000

When the input is zero, choose initial conditions $x_1(0^+)$ and $x_2(0^+)$ such that $y(t) = Ae^{-2t}$ for $t > 0$.

Q9.99.1 MCQ 1999 1M

The system model described by the state equation given below is

$$\dot{x} = \begin{bmatrix}0 & 1\\ 2 & -3\end{bmatrix} x + \begin{bmatrix}0\\ 1\end{bmatrix} u$$

$$y = \begin{bmatrix}1 & 1\end{bmatrix} x$$

A. Controllable and Observable.

B. Controllable but not Observable.

C. Observable but not Controllable.

D. Neither Controllable nor Observable.

Q9.97.1 MCQ 1997 2M

A certain linear time invariant system having the state and the output equation given below,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}1 & -1\\ 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix} u$$

$$y = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

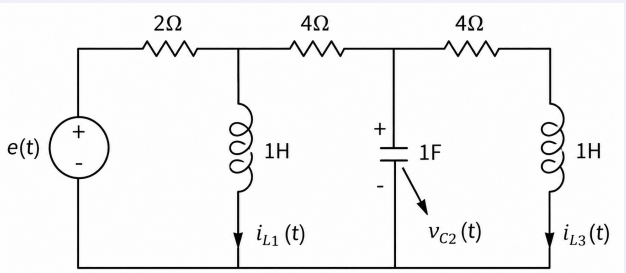
If $x_1(0) = 1, x_2(0) = -1, u(0) = 0$, then $\left.\frac{dy}{dt}\right|_{t=0}$ is

A. 1 **B.** -1 **C.** 0 **D.** None of these

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.97.2 and Q9.97.3 using the following data. These two questions together carry a total of **5 marks**.



For the circuit shown in Fig. choose state variables X_1, X_2, X_3 to be $i_{L1}(t)$, $v_{C2}(t)$, $i_{L3}(t)$

Q9.97.2 NAT 1997

Write the state equations

$$\begin{bmatrix}\dot{X}_1\\ \dot{X}_2\\ \dot{X}_3\end{bmatrix} = A\begin{bmatrix}X_1\\ X_2\\ X_3\end{bmatrix} + B\,[e(t)]$$

Q9.97.3 NAT 1997

If $e(t) = 0$, $t \geq 0$, $i_{L1}(0) = 0$, $v_{C2}(0) = 0$, $i_{L3}(0) = 1A$, then determine the total energy dissipated in the resistors in the interval $(0, \infty)$.

Q9.92.1 MSQ 1992 2M

A linear time-invariant system is described by the state variable model

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix} u$$

and

$$y = \begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

☐ **A.** The system is completely controllable.

☐ **B.** The system is not completely controllable.

☐ **C.** The system is completely observable.

☐ **D.** The system is not completely observable.

Q9.91.1 MCQ 1991 2M

A linear second order single input continuous time system is described by the following set of differential equations

$$\dot{x}_1(t) = -2x_1(t) + 4x_2(t)$$

$$\dot{x}_2(t) = 2x_1(t) - x_2(t) + u(t)$$

where $x_1(t)$ and $x_2(t)$ are the state variables and $u(t)$ is the control variable.
The system is

A. controllable and stable

B. controllable but unstable

C. uncontrollable and unstable

D. uncontrollable and stable

Q9.00.1 *	Q9.00.2 *	Q9.00.3 *
Q9.99.1 A	Q9.97.1 A	Q9.97.2 *
Q9.97.3 *	Q9.92.1 B, C	Q9.91.1 B

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q9.26.1 C	Q9.25.1 B	Q9.24.1 A, C
Q9.23.1 A	Q9.20.1 A	Q9.19.1 A
Q9.17.1 4.99 to 5.01	Q9.17.2 D	Q9.16.1 A
Q9.15.1 D	Q9.14.1 D	Q9.14.2 B
Q9.14.3 C	Q9.14.4 A	Q9.14.5 B
Q9.13.1 A	Q9.13.2 A	Q9.12.1 D
Q9.11.1 B	Q9.10.1 D	Q9.10.2 C
Q9.09.1 C	Q9.08.1 C	Q9.07.1 A
Q9.07.2 A	Q9.07.3 D	Q9.06.1 A
Q9.05.1 A, B	Q9.04.1 D	Q9.04.2 B
Q9.03.1 C	Q9.02.1 D	Q9.02.2 *
Q9.02.3 *	Q9.02.4 *	Q9.01.1 A

UNIT

X

Controllers and Compensators

Topics

1. Basics and Types of Controllers — P, PI, PD, PID
2. Lag Compensator
3. Lead Compensator
4. Lag-Lead Compensator
5. Design Problems on Controllers and Compensators

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 26	**MCQ:** 18	**MSQ:** 1	**NAT:** 7
1M: 6	**2M:** 16	**Descriptive:** 4	**Avg Weightage** ~2M	**Range:** 1M(2017)–4M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q10.26.1 MCQ 2026 2M

For the control system shown in the Figure, the transfer function of a plant,

$$G(s) = \frac{1}{(s+1)(s+2)}$$

is connected in cascade with a compensator $C(s) = K(s + \alpha)$, where K and α are positive real valued constants.
Which of the following pairs (K, α) represent the correct values for the closed loop system to have poles at $(-3 \pm j\sqrt{5})$?

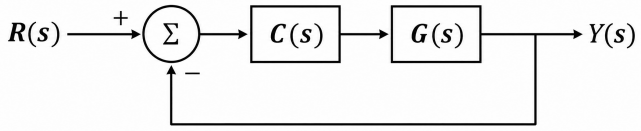


Fig. Q10.26.1

A. 2, 3 **B.** 3, 4 **C.** 2, 4 **D.** 3, 3

Q10.25.1 MCQ 2025 2M

Let $G(s) = \frac{1}{10s^2}$ be the transfer function of a second-order system. A controller $M(s)$ is connected to the system $G(s)$ in the configuration shown below.

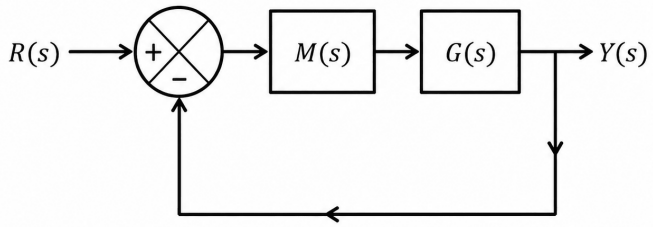


Fig. Q10.25.1

Consider the following statements.
(i) There exists no controller of the form $M(s) = \frac{K_I}{s}$, where K_I is a positive real number, such that the closed loop system is stable.
(ii) There exists at least one controller of the form $M(s) = K_P + sK_D$, where K_P and K_D are positive real numbers, such that the closed loop system is stable.
Which one of the following options is correct?

A. (i) is TRUE and (ii) is FALSE

B. (i) is FALSE and (ii) is TRUE

C. Both (i) and (ii) are FALSE

D. Both (i) and (ii) are TRUE

Q10.21.1 NAT 2021 2M

A unity feedback system that uses proportional-integral (PI) control is shown in the figure.

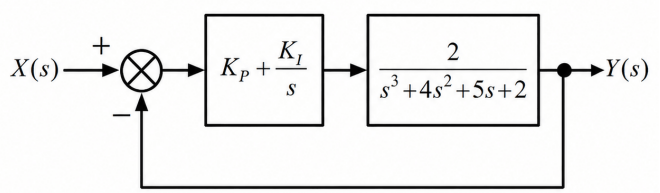


Fig. Q10.21.1

The stability of the overall system is controlled by tuning the PI control parameters K_P and K_I. The maximum value of K_I that can be chosen so as to keep the overall system stable or, in the worst case, marginally stable (rounded off to three decimal places) is

_____.

Q10.19.1 NAT 2019 2M

Consider a unity feedback system, as in the figure shown, with an integral compensator $\frac{K}{s}$ and open-loop transfer function

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

where $K > 0$. The positive value of K for which there are exactly two poles of the unity feedback system on the $j\omega$ axis is equal to

_____(*rounded off to two decimal places*).

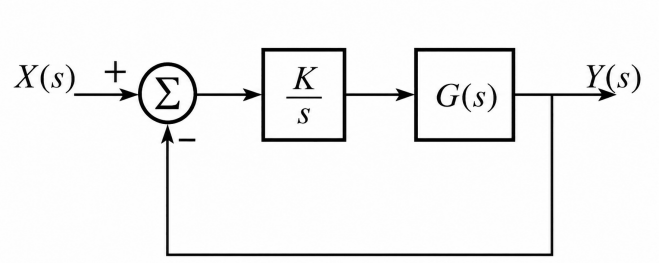


Fig. Q10.19.1

Q10.17.1 MCQ 2017 1M

Which of the following statements is incorrect?

A. Lead compensator is used to reduce the settling time.

B. Lag compensator is used to reduce the steady state error.

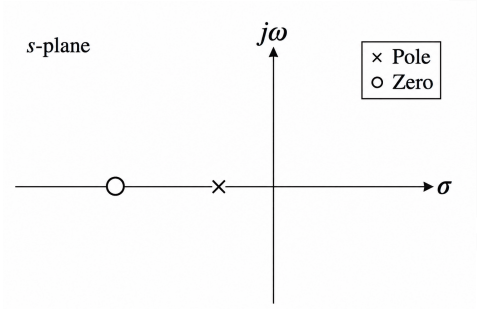
C. Lead compensator may increase the order of a system.

D. Lag compensator always stabilizes an unstable system.

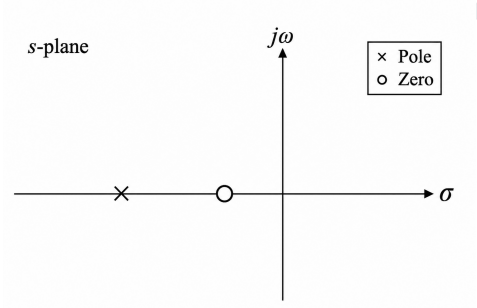
Q10.17.2 MCQ 2017 1M

Which of the following can be the pole-zero configuration of a phase-lag controller (lag compensator)?

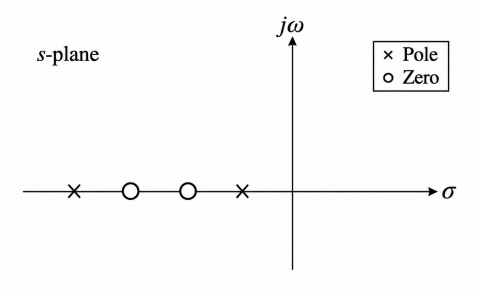
A.



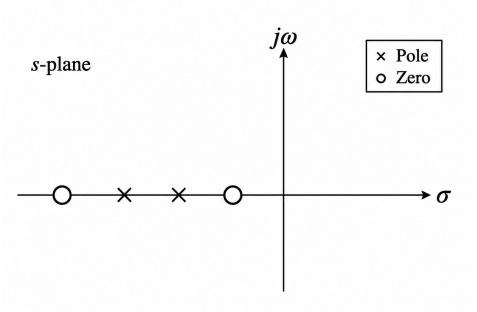
B.



C.



D.



Q10.15.1 MCQ 2015 1M

The transfer function of a first-order controller is given as

$$G_c(s) = \frac{K(s+a)}{s+b}$$

where K, a and b are positive real numbers. The condition for this controller to act as a phase lead compensator is

A. $a < b$ **B.** $a > b$ **C.** $K < ab$ **D.** $K > ab$

Q10.15.2 NAT 2015 2M

A lead compensator network includes a parallel combination of R and C in the feed-forward path. If the transfer function of the compensator is

$$G_c(s) = \frac{s+2}{s+4},$$

the value of RC is _____.

Q10.14.3 MCQ 2014 2M

For the following feedback system

$$G(s) = \frac{1}{(s+1)(s+2)}.$$

The 2%-settling time of the step response is required to be less than 2 seconds.

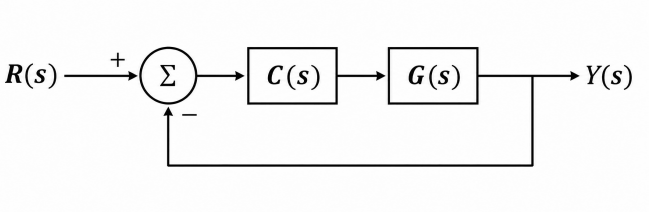


Fig. Q10.14.3

Which one of the following compensators $C(s)$ achieves this?

A. $3\left(\frac{1}{s+5}\right)$ **B.** $5\left(\frac{0.03}{s}+1\right)$

C. $2(s+4)$ **D.** $4\left(\frac{s+8}{s+3}\right)$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.12.1 and Q10.12.2 using the following data.

The transfer function of a compensator is given as

$$G_c(s) = \frac{s+a}{s+b}.$$

Q10.12.1 MCQ 2012 2M

$G_c(s)$ is a lead compensator if

A. $a = 1,\ b = 2$ **B.** $a = 3,\ b = 2$

C. $a = -3,\ b = -1$ **D.** $a = 3,\ b = 1$

Q10.12.2 MCQ 2012 2M

The phase of the above lead compensator is maximum at

A. $\sqrt{2}$ rad/s **B.** $\sqrt{3}$ rad/s

C. $\sqrt{6}$ rad/s **D.** $1/\sqrt{3}$ rad/s

Q10.10.1 MCQ 2010 2M

A unity negative feedback closed loop system has a plant with the transfer function

$$G(s) = \frac{1}{s^2+2s+2}$$

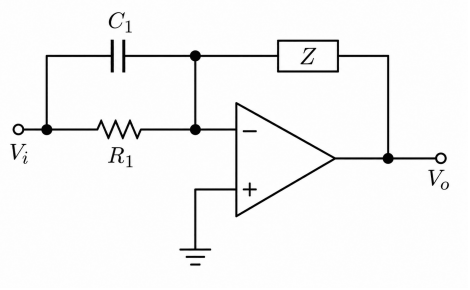
and a controller $G_c(s)$ in the feedforward path. For a unit step input, the transfer function of the controller that gives minimum steady state error is

A. $G_c(s) = \frac{s+1}{s+2}$ **B.** $G_c(s) = \frac{s+2}{s+1}$

C. $G_c(s) = \frac{(s+1)(s+4)}{(s+2)(s+3)}$ **D.** $G_c(s) = 1 + \frac{2}{s} + 3s$

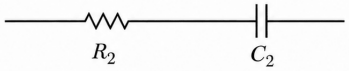
Q10.08.1 MCQ 2008 2M

Group I gives two possible choices for the impedance Z in the diagram. The circuit elements in Z satisfy the condition $R_2C_2 > R_1C_1$. The transfer function $\frac{V_o}{V_i}$ represents a kind of controller. Match the impedances in Group I with the types of controllers in Group II.



Group I

Q.



R.

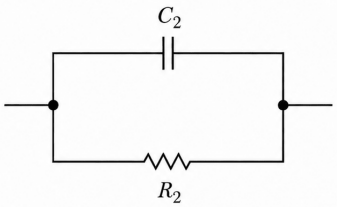


Fig. Q10.08.1

Group II

1. PID controller
2. Lead compensator
3. Lag compensator

A. Q-1, R-2 **B.** Q-1, R-3

C. Q-2, R-3 **D.** Q-3, R-2

Q10.09.1 MCQ 2009 1M

The magnitude plot of a rational transfer function $G(s)$ with real coefficients is shown below. Which of the following compensators has such a magnitude plot?

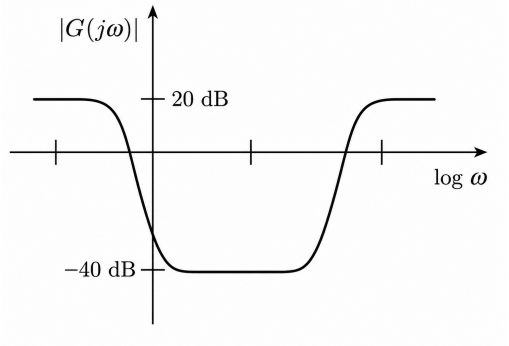


Fig. Q10.09.1

A. Lead compensator **B.** Lag compensator

C. PID compensator **D.** Lead-lag compensator

Q10.07.1 MCQ 2007 2M

A control system with a PD controller is shown in the figure. If the velocity error constant $K_v = 1000$ and the damping ratio $\zeta = 0.5$, then the values of K_P and K_D are

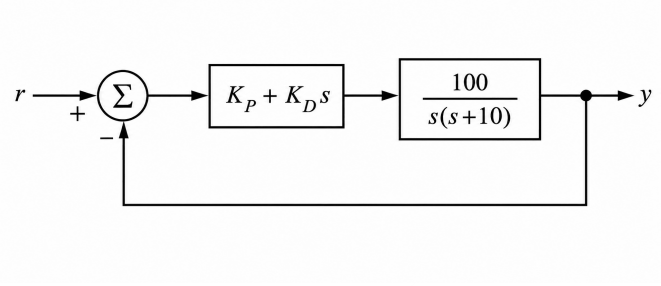


Fig. Q10.07.1

A. $K_P = 100,\ K_D = 0.09$ **B.** $K_P = 100,\ K_D = 0.9$
C. $K_P = 10,\ K_D = 0.09$ **D.** $K_P = 10,\ K_D = 0.9$

Q10.07.2 MCQ 2007 2M

The open-loop transfer function of a plant is given as

$$G(s) = \frac{1}{s^2 - 1}.$$

If the plant is operated in a unity feedback configuration, then the lead compensator that can stabilize this control system is

A. $\frac{10(s-1)}{s+2}$ **B.** $\frac{10(s+4)}{s+2}$ **C.** $\frac{10(s+2)}{s+10}$ **D.** $\frac{2(s+2)}{s+10}$

Q10.06.1 MCQ 2006 2M

The transfer function of a phase-lead compensator is given by

$$G_c(s) = \frac{1+3Ts}{1+Ts} \qquad \text{where } T > 0$$

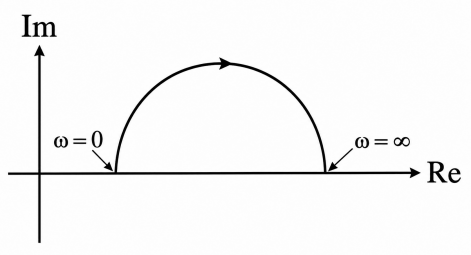
The maximum phase-shift provided by such a compensator is:

A. $\frac{\pi}{2}$ **B.** $\frac{\pi}{3}$ **C.** $\frac{\pi}{4}$ **D.** $\frac{\pi}{6}$

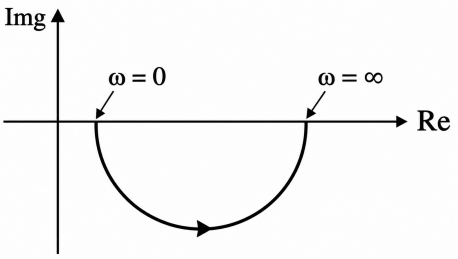
Q10.05.1 MCQ 2005 1M

Which one of the following polar diagrams corresponds to a lag network?

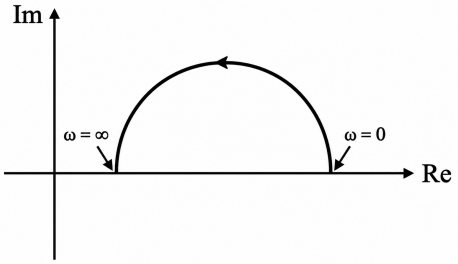
A.



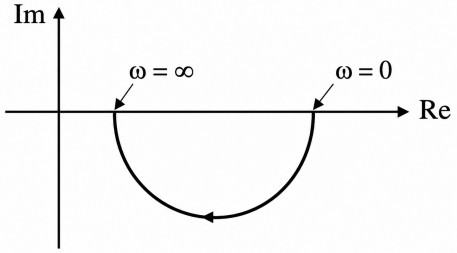
B.



C.



D.



Q10.05.2 MCQ 2005 2M

A double integrator plant,

$$G(s) = \frac{K}{s^2}, \qquad H(s) = 1$$

is to be compensated to achieve the damping ratio $\xi = 0.5$, and an undamped natural frequency, $\omega_n = 5$ rad/s. Which one of the following compensator $G_c(s)$ will be suitable?

A. $\frac{s+3}{s+9.9}$ **B.** $\frac{s+9.9}{s+3}$ **C.** $\frac{s-6}{s+8.33}$ **D.** $\frac{s+6}{s}$

Q10.03.1 MCQ 2003 1M

A PD controller is used to compensate a system. Compared to the uncompensated system, the compensated system has

A. a higher type number

B. reduced damping

C. higher noise amplification

D. larger transient overshoot

Common Data Questions

The next 3 questions share the data below.

Answer Questions Q10.02.1, Q10.02.2 and Q10.02.3 using the following data. These three questions together carry a total of **5 marks**.

A unity feedback system has the plant transfer function

$$G_p(s) = \frac{1}{(s+1)(2s+1)}$$

Q10.02.1 NAT 2002

Determine the frequency at which the plant has a phase lag of 90°.

Q10.02.2 NAT 2002

An integral controller with transfer function

$$G_c(s) = \frac{k}{s}$$

is placed in the feed-forward path of the feedback system. Find the value of k such that the compensated system has an open-loop gain margin of 2.5.

Q10.02.3 NAT 2002

Determine the steady state errors of the compensated system to unit step and unit ramp inputs.

Common Data Questions

The next 3 questions share the data below.

Answer Questions Q10.01.1, Q10.01.2 and Q10.01.3 using the following data. These three questions together carry a total of **5 marks**.

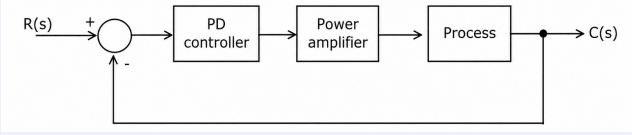
Questions corresponding to Q10.01.1 and Q10.01.3 are available at Q5.01.2 and Q5.01.3 respectively.

For the feedback control system shown in the below figure, the process transfer function is

$$G_p(s) = \frac{1}{s(s+1)}$$

and the amplification factor of the power amplifier is $K \geq 0$.

The design specifications required for the system are a time constant of 1 sec and a damping ratio of 0.707.



Q10.01.2 NAT 2001

Write down the required characteristic equation for the system. Hence determine the PD controller transfer function $G_0(s)$ when $K = 1$.

Q10.92.1 MSQ 1992 2M

A process with open-loop model

$$G(s) = \frac{Ke^{-sT_d}}{s\tau + 1}$$

is controlled by a PID controller. For this process

- **A.** the integral mode improves transient performance.
- **B.** the integral mode improves steady state performance.
- **C.** the derivative mode improves transient performance.
- **D.** the derivative mode improves steady state performance.

Q10.90.1 MCQ 1990 2M

The transfer function of a simple RC network function as a controller is

$$G_c(s) = \frac{s + Z_1}{s + P_1}$$

The condition for the RC network to act as a phase lead compensator is

A. $P_1 < Z_1$ **B.** $P_1 = 0$ **C.** $P_1 = Z_1$ **D.** $P_1 > Z_1$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q10.26.1 B	Q10.25.1 D	Q10.21.1 3.125 to 3.125
Q10.19.1 5.99 to 6.01	Q10.17.1 D	Q10.17.2 A
Q10.15.1 A	Q10.15.2 0.5 to 0.5	Q10.14.3 C
Q10.12.1 A	Q10.12.2 A	Q10.10.1 D
Q10.08.1 B	Q10.09.1 *	Q10.07.1 B
Q10.07.2 C	Q10.06.1 D	Q10.05.1 D

Q10.05.2 A	Q10.03.1 C	Q10.02.1 *
Q10.02.2 *	Q10.02.3 *	Q10.01.2 *
Q10.92.1 B, C	Q10.90.1 D	

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